Proétale cohomology in Lean

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Chapter 1

Local structures

This chapter serves for the first half of the commutative algebra preparation of the remaining part of the paper. The key result is

1. Theorem 1.138 expressing that For any ring A, there exists an ind-étale faithfully flat A-algebra D with D w-contractible. That is to say, every affine scheme has a w-contractible cover in the "pro-(etale site)".

1.1 Preliminaries

Definition 1.1 (Extremally disconnected space)**.** A topological space X is *extremally disconnected* if the closure of every open subset is open.

The reason that we are interested in extremally disconnected space is the following theorem.

Theorem 1.2. *Let X be a quasi-compact Hausdorff topological space. Then X is extremally disconnected if and only if it is a projective object in the category of quasi-compact Hausdorff spaces, i.e., for every continuous surjection $f : Y \to Z$ of quasi-compact Hausdorff spaces and every continuous map $g : X \to Z$, there exists a continuous map $h : X \to Y$ such that $f \circ h = g$.*

Proof. Omitted. □

Definition 1.3 (Stone-Čech compactification)**.** Let X be a topological space. The *Stone-Čech compactification* of X is the profinite space $\beta(X)$ such that

1. X is dense in $\beta(X)$;
2. every continuous map $f : X \to Y$ to a quasi-compact Hausdorff space Y extends uniquely to a continuous map $\beta(f) : \beta(X) \to Y$.

We denote the Stone-Čech compactification of X by $\beta(X)$.

Theorem 1.4. *Let X be a topological space. Then the Stone-Čech compactification $\beta(X)$ is extremally disconnected.*

Proof. Omitted. □

Proposition 1.5 ([Sta18, Tag 090D])**.** *Let X be a quasi-compact Hausdorff space. There exists a continuous surjection $X' \to X$ with X' quasi-compact, Hausdorff, and extremally disconnected.*

Proof. Let $Y = X$ but endowed with the discrete topology. Let $X' = \beta(Y)$, which is extremally disconnected by Theorem 1.4. The continuous map $Y \to X$ factors as $Y \to X' \to X$. $\square$

We need the following lemmas about connected components. jiedong: This is WRONG. In general, $\pi_0(X) \times \pi_0(Y)$ is continuous and bijective, but not necessarily homeomorphic.

Lemma 1.6. *Let X, Y be topological spaces. Then we have $\pi_0(X \times Y) \simeq \pi_0(X) \times \pi_0(Y)$ as topological spaces.*

Proof. Let x, y be points of X, Y, respectively. To show the above map is bijective, it suffices to show that the connected component of (x, y) in $X \times Y$ is the product of the connected component of x in X and the connected component of y in Y. We know that the product of connected subsets is still connected (`IsPreconnected.prod`). The connected component of (x, y) cannot be larger because its projections to X and Y are also connected.

To show that the bijective map is homeomorphic, we notice that the topology on both side are generated by the basis of form image $U \times$ image V. $\square$

Lemma 1.7. *Let X be a totally disconnected topological space. Then $X \simeq \pi_0(X)$ as topological spaces.*

Proof. Use the universal property of connected components. $\square$

Lemma 1.8. *Let $X = \lim_i X_i$ be a limit of topological spaces over some index category I and let $s \subseteq X$ be a subset of X. Denote by s_i the image of s in X_i. Then $\overline{s} = \bigcap_i \pi_i^{-1}(\overline{s_i})$, where $\pi_i : X \to X_i$ is the projection map.*

Proof. Omitted. $\square$

Lemma 1.9. *Let $X = \lim_i X_i$ be a limit of topological spaces over some index category I and let $s \subseteq X$ be a subset of X. Denote by s_i the image of s in X_i. Then for any morphism $f : i \to j$ in I with transition map $\varphi_f : X_i \to X_j$, we have $\varphi_f(\overline{s_i}) \subseteq \overline{s_j}$.*

Proof. Omitted. $\square$

Lemma 1.10. *Let $f \colon X \to Y$ be an embedding of topological spaces. Then f preserves and reflects specializations, i.e. for $a, b \in X$, a specializes to b if and only if $f(a)$ specializes to $f(b)$.*

Proof. The only if direction holds for any continuous map. Conversely, if $f(a)$ specializes to $f(b)$, then $b \in f^{-1}(\overline{\{f(a)\}}) = \overline{\{a\}}$. $\square$

Definition 1.11. Let X be a topological space and $Z \subseteq X$ a subset. The set

$$Z^g = \{x \in X \mid \exists z \in Z, x \rightsquigarrow z\}$$

is called the *generalization of Z in X*.

Lemma 1.12. *Let $Z \subseteq X$ be a subset of a topological space. Then Z^g is closed under generalization.*

Proof. This is immediate from the definition. $\square$

The following pure topological lemmas will be used in Lemmas 1.73 and 1.126.

Definition 1.13 ([Sta18, Tag 08ZL])**.** Let X be a topological space. Let $\pi_0(X)$ be the set of connected components of X. Let $X \to \pi_0(X)$ be the map which sends $x \in X$ to the connected component of X passing through x. Endow $\pi_0(X)$ with the quotient topology. Then any continuous map $X \to Y$ from X to a totally disconnected space Y factors through $\pi_0(X)$.

Lemma 1.14. *Let X be a topological space. Let $\pi_0(X)$ be the topological space of connected components of X. Let Y be a totally disconnected space. If every fiber of $X \to Y$ is connected, then the map $\pi_0(X) \to Y$ is injective.*

Proof. Omitted. □

Lemma 1.15 ([Sta18, Tag 005F])**.** *Let X be a topological space. Assume X is quasi-compact, quasi-separated and prespectral. Then for any $x \in X$ the connected component of X containing x is the intersection of all open and closed subsets of X containing x.*

Proof. Let T be the connected component containing x. Let $S = \bigcap_{\alpha \in A} Z_\alpha$ be the intersection of all open and closed subsets Z_α of X containing x. Note that S is closed in X. Note that any finite intersection of Z_α's is a Z_α. Because T is connected and $x \in T$ we have $T \subset Z_\alpha$ for every Z_α, thus $T \subset S$. It suffices to show that S is connected. If not, then there exists a disjoint union decomposition $S = B \amalg C$ with B and C open and closed in S. In particular, B and C are closed in X, and so quasi-compact by assumption (1). By assumption (2) there exist quasi-compact opens $U, V \subset X$ with $B = S \cap U$ and $C = S \cap V$. Then $U \cap V \cap S = \emptyset$. Hence $\bigcap_\alpha U \cap V \cap Z_\alpha = \emptyset$. By assumption (3) the intersection $U \cap V$ is quasi-compact. Hence for some $\alpha' \in A$ we have $U \cap V \cap Z_{\alpha'} = \emptyset$. Since $X - (U \cup V)$ is disjoint from S and closed in X hence quasi-compact, we can use the same argument to see that $Z_{\alpha''} \subset U \cup V$ for some $\alpha'' \in A$. Then $Z_\alpha = Z_{\alpha'} \cap Z_{\alpha''}$ is contained in $U \cup V$ and disjoint from $U \cap V$. Hence $Z_\alpha = U \cap Z_\alpha \amalg V \cap Z_\alpha$ is a decomposition into two open pieces, hence $U \cap Z_\alpha$ and $V \cap Z_\alpha$ are open and closed in X. Thus, if $x \in B$ say, then we see that $S \subset U \cap Z_\alpha$ and we conclude that $C = \emptyset$. □

Lemma 1.16 ([Sta18, Tag 04PL])**.** *Let X be a topological space. Assume X is quasi-compact, quasi-separated and prespectral. Then for a subset $T \subset X$ the following are equivalent:*

1. *T is an intersection of open and closed subsets of X, and*
2. *T is closed in X and is a union of connected components of X.*

Proof. It is clear that (a) implies (b). Assume (b). Let $x \in X$, $x \notin T$. Let $x \in C \subset X$ be the connected component of X containing x. By Lemma 1.15 we see that $C = \bigcap V_\alpha$ is the intersection of all open and closed subsets V_α of X which contain C. In particular, any pairwise intersection $V_\alpha \cap V_\beta$ occurs as a V_α i.e. is still open and closed. As T is a union of connected components of X we see that $C \cap T = \emptyset$. Hence $T \cap \bigcap V_\alpha = \emptyset$. Since T is quasi-compact as a closed subset of a quasi-compact space, we deduce that $T \cap V_\alpha = \emptyset$ for some α. For this α we see that $U_\alpha = X - V_\alpha$ is an open and closed subset of X which contains T and not x. The lemma follows. □

Lemma 1.17 ([Sta18, Tag 0906])**.** *Let X be a spectral space. Then $\pi_0(X)$ is a profinite space.*

Proof. We will show that $\pi_0(X)$ is Hausdorff, quasi-compact, and totally disconnected. By Lemma 1.15, every connected component of X is the intersection of the open and closed subsets containing it. Since $\pi_0(X)$ is the image of a quasi-compact space, it is also quasi-compact. It is totally disconnected by construction. Let $C, D \subset X$ be distinct connected components of X. Write $C = \bigcap_\alpha U_\alpha$ as the intersection of the open and closed subsets of X containing C. Any finite intersection of U_α's is another U_α. Since $\bigcap_\alpha U_\alpha \cap D = \emptyset$ we conclude that $U_\alpha \cap D = \emptyset$ for some α (connected components are closed, thus quasi-compact). Since U_α is open and closed, it is the union of the connected components it contains, i.e., U_α is the inverse image of some open and closed subset $V_\alpha \subset \pi_0(X)$. This proves that the points corresponding to C and D are contained in disjoint open subsets, i.e., $\pi_0(X)$ is Hausdorff. □

Lemma 1.18. *Let X be a spectral space. Let $Y \subset X$ be a closed subspace. Then Y is spectral.*

Proof. The only remaining part in Lean is quasi-separatedness. Suppose that V is open compact in Y, we show that there exists open compact U in X such that $V = U \cap Y$. We can find U' in X such that $V = U' \cap Y$. Write $U' = \bigcup_{i\in I} U_i$ as a union of compact opens, then a finite subset $i \in I_0$ of U_i already covers V. Take $U := \bigcup_{i\in I\_0} U_i$ as desired.

For any compact opens V_1, $V_2 \subset Y$, there exists compact open $U_1, U_2 \subset X$ s.t. $V_i = U_i \cap Y_i$. Then $V_1 \cap V_2 = (U_1 \cap U_2) \cap Y$, is still open compact in Y. Here $U_1 \cap U_2$ is quasi-compact since X is quasi-separated. □

Lemma 1.19. *Let X and Y be topological spaces. Let $(x, y) \in X \times Y$ be a point. Then*

$$\overline{\{(x, y)\}} = \overline{\{x\}} \times \overline{\{y\}}.$$

Proof. It is clear that $\overline{\{(x, y)\}} \subseteq \overline{\{x\}} \times \overline{\{y\}}$.

If $z \notin \overline{\{(x, y)\}}$, there exists opens $U \subset X$, $V \subset Y$ such that $z \notin U \times V$, thus $(x, y) \notin U \times V$. Either $x \notin U$ or $y \notin V$, suppose $x \notin U$. Then $U \cap \overline{\{x\}} = \emptyset$, thus $U \times V \cap \overline{\{x\}} \times \overline{\{y\}} = \emptyset$. This shows that $\overline{\{x\}} \times \overline{\{y\}} \subseteq \overline{\{(x, y)\}}$. □

Lemma 1.20 ([Sta18, Tag 0907])**.** *Let X and Y be spectral spaces. Then the product $X \times Y$ is spectral.*

Proof. Let X, Y be spectral spaces. Denote $p : X \times Y \to X$ and $q : X \times Y \to Y$ the projections. We first show that $X \times Y$ is sober. Let $Z \subset X \times Y$ be a closed irreducible subset. Then $p(Z) \subset X$ is irreducible and $q(Z) \subset Y$ is irreducible. Let $x \in X$ be the generic point of the closure of $p(Z)$ and let $y \in Y$ be the generic point of the closure of $q(Z)$. If $(x, y) \notin Z$, then there exist opens $x \in U \subset X$, $y \in V \subset Y$ such that $Z \cap (U \times V) = \emptyset$. Hence Z is contained in $(X - U) \times Y \cup X \times (Y - V)$. Since Z is irreducible, we see that either $Z \subset (X - U) \times Y$ or $Z \subset X \times (Y - V)$. In the first case $p(Z) \subset (X - U)$ and in the second case $q(Z) \subset (Y - V)$. Both cases are absurd as x is in the closure of $p(Z)$ and y is in the closure of $q(Z)$. Thus we conclude that $(x, y) \in Z$, which means that (x, y) is the generic point for Z (Lemma 1.19).

A basis of the topology of $X \times Y$ consists of opens of the form $U \times V$ with $U \subset X$ and $V \subset Y$ quasi-compact open (here we use that X and Y are spectral). Then $U \times V$ is quasi-compact as the product of quasi-compact spaces is quasi-compact. Moreover, any quasi-compact open of $X \times Y$ is a finite union of such quasi-compact rectangles $U \times V$. It follows that the intersection of two such is again quasi-compact (since X and Y are spectral). This concludes the proof. □

Lemma 1.21 ([Sta18, Tag 08ZE])**.** *Let X be a topological space. Then X is Hausdorff if and only if the diagonal map $\Delta : X \to X \times X$ is a closed embedding.*

Proof. Omitted. □

Lemma 1.22 ([Sta18, Tag 08ZH])**.** *Let X, Y, and S be topological spaces. Assume that S is Hausdorff. Let $f : X \to S$ and $g : Y \to S$ be continuous maps. The natural map $X \times_S Y \to X \to Y$ is a closed embedding.*

Proof. $X \times_S Y$ is the inverse image of the image $\Delta(S)$ of the diagonal map $\Delta : S \to S \times S$. It follows from Lemma 1.21 that $\Delta(S)$ is a closed subset. □

Lemma 1.23. *Let X and Y be topological spaces. If X is spectral (resp. T0, Hausdorff, Sober, quasi-separated, quasi-compact, prespectral) and there is a homeomorphism $X \cong Y$, then Y is spectral (resp. T0, Hausdorff, Sober, quasi-separated, quasi-compact, prespectral).*

Proof. Omitted. □

Lemma 1.24. *Let X and Y be spectral spaces. Let S be a Hausdorff topological space. Let $f : X \to S$ and $g : Y \to S$ be continuous maps. Then the fiber product $X \times_S Y$ is spectral.*

Proof. Note that $X \times_S Y$ is a closed subspace of $X \times Y$ by Lemma 1.22. (For subspace, see `TopCat.pullbackIsoProdSubtype`.) Since $X \times Y$ is spectral (Lemma 1.20) it follows that $X \times_S Y$ is spectral (Lemma 1.18). Note that we also used Lemma 1.23 to translate between categorical constructions and non-categorical constructions. □

Lemma 1.25. *Let X be a profinite space. Then X is spectral.*

Proof. Since R1 implies Sober, Hausdorff implies quasi-separated, it only remains to show that X is prespectral. Write X as an inverse limit of finite discrete topological spaces. The preimage of singletons in every finite discrete space are open and closed subsets and forms a topological basis for the topology on X. Thus X is prespectral. □

Lemma 1.26 ([Sta18, Tag 096C, first part])**.** *Let X be a spectral space. Let*

$$\begin{array}{ccc} Y & \longrightarrow & T \\ \downarrow & & \downarrow \\ X & \longrightarrow & \pi_0(X) \end{array}$$

be a cartesian diagram in the category of topological spaces with T profinite. Then Y is spectral and the canonical map $\pi_0(Y) \to T$ defined in Definition 1.13 is an isomorphism.

Proof. By Lemma 1.25, T is spectral. By Lemma 1.17 $\pi_0(X)$ is Hausdorff. By Lemma 1.24, Y is spectral. Let $Y \to \pi_0(Y) \to T$ be the canonical factorization in Definition 1.13. It is clear that $\pi_0(Y) \to T$ is surjective. The fibres of $Y \to T$ are homeomorphic to the fibres of $X \to \pi_0(X)$. Hence these fibres are connected. It follows that $\pi_0(Y) \to T$ is injective by Lemma 1.14. We conclude that $\pi_0(Y) \to T$ is a homeomorphism since it is a bijective map from a quasi-compact space to a Hausdorff space by `isHomeomorph_iff_continuous_bijective`. □

The following pure algebraic lemma will be used in Lemma 1.79.

Lemma 1.27 ([Sta18, Tag 00EA, (1) + (2)(a)])**.** *Let $A \to B$ be a ring map which has going down. Let $\mathfrak{p} \subset A$ be a prime ideal and let $\mathfrak{q} \subset B$ be a prime ideal lying over $\mathfrak{p}$. Assume that and there is at most one prime of B above every prime of A. Then the natural map $B_{\mathfrak{p}B} \to B_{\mathfrak{q}}$ is an isomorphism.*

Proof. It suffices to show that all the elements of $B - \mathfrak{q}$ are all invertible in $B_{\mathfrak{p}B}$. Suppose the contrary, let $x \in B - \mathfrak{q}$ be an element not invertible in $B_{\mathfrak{p}B}$. We can find a prime ideal $\mathfrak{q}' \subseteq \mathfrak{p}B$ of B containing x. By going down there exists a prime $\mathfrak{q}'' \subseteq \mathfrak{q}$ lying over $\mathfrak{p}' = A \cap \mathfrak{q}'$. By the uniqueness of primes lying over $\mathfrak{p}'$ we see that $\mathfrak{q}' = \mathfrak{q}''$, which contradicts the fact that $x \notin \mathfrak{q}$. □

Definition 1.28 (Local isomorphisms, [Sta18, Tag 096E (1)])**.** We say $A \to B$ is a *local isomorphism* if for every prime $\mathfrak{q} \subset B$ there exists a $g \in B$, $g \notin \mathfrak{q}$ such that $A \to B_g$ induces an open immersion $\mathrm{Spec}(B_g) \to \mathrm{Spec}(A)$.

Definition 1.29 (Identify local rings, [Sta18, Tag 096E (2)]). A ring map $A \to B$ *identifies local rings* if for every prime $\mathfrak{q} \subset B$ the canonical map $A_{\varphi^{-1}(\mathfrak{q})} \to B_{\mathfrak{q}}$ is an isomorphism.

Lemma 1.30. *The map $A \to \prod_{i=1}^{n} A_i$ identifies local rings, if $A \to A_i$ identifies local rings for all i.*

Proof. Omitted. $\square$

Lemma 1.31. *Let $A \to B$ and $B \to C$ be ring maps that identify local rings. Then the composition $A \to C$ also identifies local rings.*

Proof. Omitted. $\square$

Lemma 1.32 ([Sta18, Tag 096D]). *Let A be a ring and $B \to C$ be an A-algebra map. Suppose both $A \to B$ and $A \to C$ are bijective on stalks. Then $B \to C$ is bijective on stalks.*

Proof. Holds because the same holds for bijective maps of sets. $\square$

Definition 1.33 ([Sta18, Tag 096L]). Let A be a ring and $X = \operatorname{Spec}(A)$. Let ILR_A denote the category of A-algebras B for which $A \to B$ identifies local rings, and let Top_X denote the category of topological spaces over X.

Define the functor

$$F\colon \mathrm{ILR}_A \longrightarrow \mathrm{Top}_X$$

by sending B to $\operatorname{Spec}(B)$.

Lemma 1.34. *[Sta18, Tag 096L] Let A be a ring. Set $X = \operatorname{Spec}(A)$. The functor F constructed in Definition 1.33*

$$B \longmapsto \operatorname{Spec}(B)$$

from the category of A-algebras B such that $A \to B$ identifies local rings to the category of topological spaces over X is fully faithful.

Proof. The functor F is a composition of two functors: 1. The fully faithful functor from the category of A-algebras B for which $A \to B$ identifies local rings to the category of ringed spaces $(Y, \mathcal{O}_Y)$ over $X = \operatorname{Spec}(A)$ satisfying $\mathcal{O}_Y = p^{-1}\mathcal{O}_X$, where $p : Y \to X$ is the structure map. 2. The functor sending a ringed space to its underlying topological space, and a morphism $(f, f^{\#})$ to f.

The second functor is fully faithful because $f^{\#}$ is always an isomorphism, being given by the canonical identification $f^{-1}\mathcal{O}_Y \cong f^{-1}p^{-1}\mathcal{O}_X = q^{-1}\mathcal{O}_X \cong \mathcal{O}_Z$, where $p : Y \to X$ and $q : Z \to X$ are the structure maps. $\square$

Definition 1.35. Let A be a ring and $X = \operatorname{Spec}(A)$. Let $A \to B$ and $A \to C$ be two A-algebras that identifies local rings. As a consequence of Lemma 1.34, we have a bijective map

$$F : \operatorname{Hom}_{A\text{-alg}}(B, C) \to \operatorname{Hom}_{\text{-}\mathrm{Top}_X}(\operatorname{Spec}(C), \operatorname{Spec}(B)).$$

sending $f : B \to C$ to the continuous map $\operatorname{Spec}(f) : \operatorname{Spec}(C) \to \operatorname{Spec}(B)$ induced by f.

Lemma 1.36. *Let R be a commutative semiring, M a finitely generated R-module, $S \subseteq R$ a submonoid and $g\colon M \to M'$ a localization of M at S. Then M' is the trivial module if and only if there exists $t \in S$ such that $t \cdot m = 0$ for all $m \in M$.*

Proof. Omitted. $\square$

Lemma 1.37. *Let $A = \operatorname{colim}_i A_i$ be a filtered colimit of A-algebras and let B be an étale A-algebra. Then there exists an i and an étale A_i-algebra B' such that*

$$\begin{tikzcd} A_i \arrow[r] \arrow[d] & B' \arrow[d] \\ A \arrow[r] & B \end{tikzcd}$$

is a pushout diagram.

Proof. This follows because every étale algebra is of finite presentation and by possibly enlarging i we can ensure that B' is étale over A_i. The latter step uses Lemma 1.36 to find a single index i whose stage trivialises the relevant cotangent / differentials module. □

1.2 ind-Zariski maps

Definition 1.38 (Ind-Zariski, [Sta18, Tag 096N])**.** An R-algebra S is called a *ind-Zariski* if S can be written as a filtered colimit $S \simeq \operatorname{colim} S_i$ with each $R \to S_i$ a local isomorphism. A ring homomorphism $f : R \to S$ is called a *ind-Zariski* if S is ind-Zariski as an R-algebra via f.

Remark 1.39. The definition of ind-Zariski is slightly more general from [BS15, Definition 2.2.1(iv)], where they defined as a filtered colimit of finite products of principal localizations.

Lemma 1.40 ([Sta18, Tag 096T])**.** *Let $A \to B$ be an ind-Zariski ring map. Then it identifies local rings, i.e. for every prime $\mathfrak{q} \subset B$ the canonical map $A_{\varphi^{-1}(\mathfrak{q})} \to B_{\mathfrak{q}}$ is an isomorphism.*

Proof. Omitted. □

Lemma 1.41. *Let $A \to B$ be an ind-Zariski ring map. Then $A \to B$ is flat.*

Proof. An ind-Zariski ring map is a filtered colimit of local isomorphisms. A local isomorphism is flat. □

Lemma 1.42. *A filtered colimit of ind-Zariski algebras is ind-Zariski.*

Proof. This follows from general theory, because local isomorphisms are of finite presentation. □

Lemma 1.43. *A finite product of ind-Zariski algebras is ind-Zariski.*

Proof. This is the consequence of the more general fact that if a property p of ring maps is stable under finite products, then Ind-p is also stable under finite products since finite product of filtered index category is again filtered.

It suffices to show that being local isomorphism is stable under finite products. Suppose $A \to B_i$ is a local isomorphism for each i for finitely many i's. Every prime $\mathfrak{q}$ in $\prod_i B_i$ is an extension prime ideal of one of the factors, say B_0. Since there exists $f_0 \in B_0$, $f_0 \notin \mathfrak{q}$ such that $A \to (B_0)_{f_0}$ induces an open immersion, localize $\prod_i B_i$ at the corresponding idempotent times this element f_0 to obtain the desired result. □

Lemma 1.44. *Let $A \to B$ and $B \to C$ be ind-Zariski ring maps. Then the composition $A \to C$ is also ind-Zariski.*

Proof. Omitted. □

Lemma 1.45. *Let A be a ring and $S \subseteq A$ a multiplicative subset. Then $S^{-1}A$ is ind-Zariski over A.*

Proof. $S^{-1}A = \operatorname{colim}_{f \in S} A_f$ and $A \to A_f$ is a local isomorphism. □

1.3 Henselian objects and Henselisation

In this section we define a generalisation of the notion of a Henselian ring to an arbitrary category. Let $\mathcal{C}$ be a category and P a property of morphisms.

Definition 1.46 (Henselian)**.** A morphism $f\colon X \to Y$ is *P-Henselian*, if for every factorisation

$$\begin{tikzcd} X \arrow[r, "u"] \arrow[dr, "f"'] & Z \arrow[d, "v"] \\ & Y, \end{tikzcd}$$

with u satisfying P, there exists a retraction of u.

Example 1.47. A local ring $(R, \mathfrak{m}, \kappa)$ is Henselian if and only if the projection $R \to \kappa$ is étale-Henselian.

Proof. By [Sta18, Tag 04GG], R is Henselian if and only if for any étale ring map $R \to S$ and prime $\mathfrak{q}$ of S lying over $\mathfrak{m}$ with $\kappa = \kappa(\mathfrak{q})$, there exists a retraction $S \to R$ of $R \to S$.

Suppose R is Henselian and let $R \to S \to \kappa$ be a factorisation of $R \to \kappa$ with $R \to S$ étale. Then $\mathfrak{q} = \ker(S \to \kappa)$ is a prime ideal of S lying above $\mathfrak{m}$. Hence we obtain a commutative diagram

$$\begin{tikzcd} R \arrow[r] \arrow[d] & S \arrow[r] \arrow[d] & \kappa \\ \kappa \arrow[r, dashed] & \kappa(\mathfrak{q}), \arrow[ur, dashed] & \end{tikzcd}$$

where the composition of the dashed arrows is the identity of κ. Hence $\kappa(\mathfrak{q}) = \kappa$ and by assumption, $R \to S$ has a retraction.

Now assume that $R \to \kappa$ is étale-Henselian and let $R \to S$ be étale and $\mathfrak{q}$ be a prime ideal of S with $\kappa(\mathfrak{q}) = \kappa$. Hence the composition $R \to S \to \kappa(\mathfrak{q}) = \kappa$ is a factorisation of $R \to \kappa$. Thus $R \to S$ has a retraction. □

Definition 1.48 (Strictly Henselian local ring)**.** A local ring $(R, \mathfrak{m}, \kappa)$ is *strictly henselian* if the composition $R \to \kappa \to \kappa^{\text{sep}}$ is étale-Henselian for a separable closure κ^{sep}.

The following example shows that our definition agrees with the one in [Sta18, Tag 04GF].

Lemma 1.49. *A local ring $(R, \mathfrak{m}, \kappa)$ is strictly Henselian if and only if it is Henselian and κ is separably closed.*

Proof. TBA. □

Proposition 1.50. *Let $A, \mathfrak{m}$ be a strictly henselian local ring. Let $f : A \to B$ a étale ring map with $\mathfrak{n}$ a maximal ideal of B lying over $\mathfrak{m}$. Then there exists a retraction $s : B \to A$ of $A \to B$ such that $\mathfrak{n} = s^{-1}(\mathfrak{m})$.*

Proof. Since $B/\mathfrak{m}B$ is an étale algebra over the residue field $k = A/\mathfrak{m}$, it splits as $B/\mathfrak{m}B \cong k_1 \times \cdots \times k_n$ for finitely many finite separable extensions k_i of k. Thus $B/\mathfrak{n}$, as a quotient of $B/\mathfrak{m}B$, is also a finite separable extension of k. Choose an inclusion $B/\mathfrak{n} \hookrightarrow k^{\text{sep}}$. Then we have a commutative diagram

$$\begin{tikzcd} A \arrow[r] \arrow[d] & B \arrow[d] \arrow[dr] & \\ k \arrow[r] & B/\mathfrak{n} \arrow[r] & k^{\text{sep}}. \end{tikzcd}$$

Since $A \to k^{\mathrm{sep}}$ is étale-Henselian, there exists a retraction $r : B \to A$ of $A \to B$. But this retraction may not satisfy $\mathfrak{n} = r^{-1}(\mathfrak{m})$.

To solve this, notice that there are only finitely many prime ideals $\mathfrak{n}_0 = \mathfrak{n}, \mathfrak{n}_1, \cdots, \mathfrak{n}_k$ of B lying over $\mathfrak{m}$ and they are all maximal ideals. (directly by the fact that $B/\mathfrak{m}B \cong k_1 \times \cdots \times k_n$). Since $\bigcap_{i \neq 0} \mathfrak{n}_i \not\subseteq \mathfrak{n}$ (otherwise one of $\mathfrak{n}_i \subseteq \mathfrak{n}$ which is impossible), we can find element $b \in \mathfrak{n}_i$ for every $i \neq 0$ that does not fall in $\mathfrak{n}$. We may replace $(B, \mathfrak{n})$ by $(B_b, \mathfrak{n}B_b)$, then B is still étale over A and $\mathfrak{n}$ is the only prime ideal of B lying over $\mathfrak{m}$. Then the section $s_b : B_b \to A$ given by the previous paragraph satisfies $\mathfrak{n}B_b = s_b^{-1}(\mathfrak{m})$. Composition with the localization map $B \to B_b$ gives the desired section $s : B \to A$. ☐

Proposition 1.51. *Let A be a strictly Henselian local ring and $f : A \to B$ an étale ring map. Suppose $\mathfrak{n}$ is a maximal ideal of B lying over the maximal ideal $\mathfrak{m}$ of A. Then the induced map $A \to B_\mathfrak{n}$ is an isomorphism.*

Proof. Let $s : B \to A$ be the section given by Proposition 1.50 for the pair $(B, \mathfrak{n})$. Localizing at $\mathfrak{m}$, we obtain the following diagram:

$$\begin{tikzcd} A = A_\mathfrak{m} \arrow[r, "f_\mathfrak{m}"] \arrow[dr, "f'"'] & B_\mathfrak{m} \arrow[r, "s_\mathfrak{m}"] \arrow[d] & A \\ & B_\mathfrak{n} \arrow[ur, dotted, "s'"'] & \end{tikzcd}$$

The dotted arrow s' exists by the universal property of localization, since $\mathfrak{n} = s^{-1}(\mathfrak{m})$. Moreover, s' is a section of f'.

Since $f_\mathfrak{m}$ is étale and $s_\mathfrak{m}$ is a section of $f_\mathfrak{m}$, it follows that $s_\mathfrak{m}$ is also étale by the cancellation property of étale maps. Thus s', being a composition of a localization and an étale map, is flat. As a flat local ring homomorphism, s' is faithfully flat and hence injective. Since s' is also a section of f', it is surjective. Therefore, s' is an isomorphism, and the composition $A \to B_\mathfrak{n}$ is an isomorphism. ☐

1.4 Ind-etale ring maps

Definition 1.52 (Ind-étale algebras, [Sta18, Tag 097I])**.** An R-algebra S is *ind-étale* if it is a filtered colimit of étale R-algebras. We say a ring homomorphism $R \to S$ is *ind-étale* if S is ind-étale as an R-algebra via f.

Lemma 1.53 ([Sta18, Tag 0BSI])**.** *Let $A \to B$ and $B \to C$ be ring maps. If $A \to B$ and $B \to C$ are ind-étale, then $A \to C$ is ind-étale.*

Proof. Omitted. ☐

Lemma 1.54 ([Sta18, Tag 0BSH])**.** *Let $A \to B$ be an ind-étale ring map. For any ring map $A \to C$, the base change $C \to B \otimes_A C$ is ind-étale.*

Proof. Omitted. ☐

Lemma 1.55. *Let $B = \mathrm{colim}_i B_i$ be a filtered colimit of ind-étale A-algebras. Then B is an ind-étale A-algebra.*

Proof. This follows from general theory, because étale maps are of finite presentation. ☐

Lemma 1.56. *Let $A \to B$ be an ind-Zariski ring map. Then $A \to B$ is ind-étale.*

Proof. An ind-Zariski ring map is a filtered colimit of local isomorphisms. A local isomorphism is étale. ☐

1.5 w-local spaces

Definition 1.57 (w-local spaces, [BS15, Definition 2.1.1], [Sta18, Tag 096A])**.** A topological space X is *w-local* if it satisfies:

1. X is spectral,
2. every point specializes to a unique closed point,
3. the subspace X^c of closed points is closed.

Definition 1.58 (w-local morphisms, [BS15, Definition 2.1.1], [Sta18, Tag 096A])**.** Let X and Y be w-local spaces. A morphism $f : X \to Y$ is *w-local* if it is spectral and the image of closed points $f(X^c) \subseteq Y^c$.

Definition 1.59 (w-purely-local morphisms)**.** Let X and Y be w-local spaces. A morphism $f \colon X \to Y$ is *w-purely-local* if it $f^{-1}(X^c) = Y^c$.

Obviously, every w-purely-local morphism is w-local.

Lemma 1.60. *Let $f : X \to Y$ and $g : Y \to Z$ be w-local maps between w-local spaces. Then the composition $g \circ f : X \to Z$ is w-local.*

Proof. Omitted. ◻

Lemma 1.61. *Let X be w-local and $Z \subseteq X$ a subset closed under specialization. Then the generalization Z^g in X is still closed under specialization.*

Proof. Let $x \rightsquigarrow y$ in X with $x \in Z^g$. Then x specializes to some $z \in Z$, which in turn specializes to a unique closed point c in X because X is w-local. Because Z is stable under specialization, $c \in Z$. Also y specializes to a unique closed point c' in X and by uniqueness we have $c' = c \in Z$. Hence $y \in Z^g$. ◻

Lemma 1.62. *Let X be w-local and $Z \subseteq X$ a subset closed under specialization. If Z is spectral, it is w-local and the inclusion $Z \to X$ maps closed points to closed points.*

Proof. Since Z is closed under specializations, we have $Z^c = X^c \cap Z$. In particular Z^c is closed in Z and $Z \to X$ preserves closed points. By 1.10, specializations in Z are the same as specializations in X. Since Z is closed under specializations, every point in Z specializes to a unique closed point in X and therefore in Z. ◻

Lemma 1.63 ([Sta18, Tag 096B])**.** *Let X be a w-local space. Then every closed subspace $Y \subseteq X$ is w-local. Moreover, the inclusion map $Y \to X$ is w-local.*

Proof. This follows from 1.62, because closed subsets are stable under specialization. The inclusion $Y \to X$ is spectral as a closed embedding and hence w-local by the lemma. ◻

Lemma 1.64. *Let X be a profinite topological space and $x, y \in X$ be distinct. Then there exist $U, V \subseteq X$ open and closed such that $X = U \coprod V$ and $x \in U$, $y \in V$.*

Proof. Write $X = \lim_{i \in I} X_i$ as a directed limit of finite discrete spaces. Let $\pi_i \colon X \to X_i$ be the projection maps. A base of the open sets of X is given by the preimages $\pi_i^{-1}(\{z\})$ for $z \in X_i$ and $i \in I$. Every open set in this base is also closed, because its complement is a finite union of such sets.

Choose an open subset in this base containing x but not y, say $U = \pi_{i_0}^{-1}(\{z_0\})$ for some $i_0 \in I$ and $z_0 \in X_{i_0}$. Then $V = X - U$ is also open and closed, and we have $X = U \coprod V$ with $x \in U$ and $y \in V$. ◻

Lemma 1.65 ([Sta18, Tag 0905, (6) $\implies$ (1)])**.** *Let X be a spectral space such that every point is closed. Then X is profinite.*

Proof. Since every point is closed, there are no nontrivial specializations between points. By Item 2 in Lemma 1.70, all constructible subsets of X are closed. In particular, every compact open subset is also closed. Since X is spectral, thus T0, for any two points there is a quasi-compact open (also closed) containing exactly one of them. Thus X is profinite by `isTotallyDisconnected_of_isClopen_set`. $\square$

Lemma 1.66. *Let X be a w-local space. Then the set of closed points X^c of X is profinite.*

Proof. It is a closed subspace of a spectral space, hence spectral. Therefore it is profinite by 1.65 $\square$

Lemma 1.67 ([Sta18, Tag 0A31, (3) $\implies$ (4)])**.** *Let X be a spectral space and $E \subseteq X$ a quasi-compact subset. Then the generalization E^g is an intersection of quasi-compact open subsets.*

Proof. Let X be a spectral space and let W denote E^g. Since every point of W specializes to a point of E we see that an open of W which contains E is equal to W. Hence since E is quasi-compact, so is W. If $x \in X$, $x \notin W$, then $Z = \overline{\{x\}}$ is disjoint from W. Since W is quasi-compact we can find a quasi-compact open U (since there is a open basis of quasi-compact opens of X and finitely many of them would suffice to cover W) with $W \subset U$ and $U \cap Z = \emptyset$. We conclude that W is an intersection of quasi-compact open subsets. $\square$

Definition 1.68 (Constructible topology, [Sta18, Tag 08YF])**.** Let X be a topological space. The *constructible topology* on X is the topology with subbase of opens the sets U and $X - U$ for all quasi-compact opens U of X.

In other words, a subset of X is open in the constructible topology if and only if it is a union of finite intersections of sets of the form U or $X - U$ for a quasi-compact open U.

Lemma 1.69 ([Sta18, Tag 0901])**.** *Let X be a spectral space. Then X is quasi-compact in the constructible topology.*

Proof. Let $\mathfrak{B}$ be the set of subsets of X that are quasi-compact open or the complement of a quasi-compact open. By definition of the constructible topology and the Alexander subbasis theorem, it suffices to show that every covering $X = \bigcup_{i \in I} B_i$ with $B_i \in \mathfrak{B}$ has a finite subcover. Since for every $B \in \mathfrak{B}$ also $X - B \in \mathfrak{B}$, it suffices to show: If $\{B_i\}_{i \in I}$ is a family of elements of $\mathfrak{B}$ such that all finite subfamilies have non-empty intersection, then $\bigcap_{i \in I} B_i$ is non-empty.

Let $\mathcal{S}$ be the set of families in $\mathfrak{B}$ that are pairwise unequal, have non-empty finite intersections and empty intersection. For the sake of contradiction, suppose $\mathcal{S}$ is non-empty. If we equip $\mathcal{S}$ with the ordering by subset inclusion, Zorn's Lemma yields a maximal family $\{B_i\}_{i \in I} \in \mathcal{S}$.

Let $I' \subseteq I$ be the indices such that B_i is closed and set $Z = \bigcap_{i \in I'} B_i$. This is a closed and non-empty subset of X, because X is quasi-compact and all finite intersections are non-empty by assumption. Note that for any $i_1, \dots, i_n \in I$ the intersection $\bigcap_{a=1}^{n} B_{i_a}$ is quasi-compact, hence $Z \cap \bigcap_{a=1}^{n} B_{i_a}$ is non-empty by the same argument.

Suppose $Z = Z' \cup Z'$ is reducible with $Z', Z'' \subsetneq Z$ closed. Because X is a spectral space, there exist quasi-compact opens $U', U'' \subseteq X$ with $U' \cap Z'$ non-empty, $U'' \cap Z''$ non-empty and $U' \cap Z'' = \emptyset = U'' \cap Z'$. Then set $B' = X - U' \in \mathfrak{B}$ and $B'' = X - U'' \in \mathfrak{B}$. We claim that

$\{B_i\}_{i\in I}$ with B' or B'' added is still in $\mathcal{S}$. Suppose, this is not the case. Then there exist finitely many indices i_a and j_b in I such that

$$B' \cap \bigcap_{a=1}^{n} B_{i_a} = \emptyset = B'' \cap \bigcap_{b=1}^{m} B_{j_b}.$$

By construction, $Z \subseteq B' \cup B''$, so in particular

$$Z \cap \bigcap_{a=1}^{n} B_{i_a} \cap \bigcap_{b=1}^{m} B_{j_b} = \emptyset.$$

which is a contradiction. Hence $\{B_i\}_{i\in I}$ is not maximal, also a contradiction. So Z is irreducible. But for every $i \in I - I'$, the set B_i is open and hence the non-empty open $Z \cap B_i$ of Z contains the unique generic point of Z. Hence $\bigcap_{i\in I} B_i = Z \cap \bigcap_{i\in I-I'} B_i$ is non-empty, which is again a contradiction. $\square$

Lemma 1.70 ([Sta18, Tag 0903])**.** *Let X be a spectral space and $E \subseteq X$ a subset closed in the constructible topology.*

1. *If $x \in \overline{E}$, then x is the specialization of a point in E.*
2. *If E is stable under specialization, then E is closed.*

Proof. It suffices to show the first claim. Let $x \in \overline{E}$ and let $\{U_i\}_{i\in I}$ be the set of quasi-compact open neighbourhoods of x. Since X is spectral, these form a neighbourhood basis of x, so $\bigcap_{i\in I} U_i$ is the set of generalizations of x. It hence suffices to show that $\bigcap_{i\in I} U_i \cap E$ is non-empty. Indeed, as X is quasi-separated, any finite intersection of the U_i is another one and since $x \in \overline{E}$, the intersection $U_i \cap E$ is non-empty for all i. Since $U_i \cap E$ is closed in the constructible topology for all i and X is quasi-compact in the constructible topology by 1.69, the intersection $\bigcap_{i\in I} U_i \cap E$ is non-empty. $\square$

Lemma 1.71. *Let X be a w-local space and $Z \subseteq X$ a closed subset. Then Z^g is closed.*

Proof. Since Z is closed, it is quasi-compact. Hence by Lemma 1.67, we see that Z^g is closed in the constructible topology as an intersection of quasi-compact opens. Since Z is stable under specialization, the same holds for Z^g by Lemma 1.61. Thus by Lemma 1.70, the claim follows. $\square$

Lemma 1.72 ([Sta18, Tag 0968])**.** *Let X be a w-local space. Then the composition $X^c \to X \to \pi_0(X)$ is a homeomorphism, where X^c denotes the closed points of X.*

Proof. Since X^c is quasi-compact and $\pi_0(X)$ is Hausdorff by Lemma 1.17, it suffices to show the map is bijective. To show injectivity let $x, y \in X^c$ be distinct. Since X^c is profinite (Lemma 1.66), there exist open and closed subsets $U_0, V_0 \subseteq X^c$ such that $X^c = U_0 \coprod V_0$ by Lemma 1.64. Let $U = U_0^g$ and $V = V_0^g$. Because X is w-local, we get $X = U \coprod V$ as sets. By Lemma 1.71, both U and V are closed and therefore also both open. Hence x and y lie in distinct connected components of X, so the images in $\pi_0(X)$ are distinct.

Since every connected component is closed, it is quasi-compact and hence contains a closed point. Thus the map is surjective. $\square$

Lemma 1.73 ([Sta18, Tag 096C, second part])**.** *Let X be a spectral space. Let*

$$\begin{tikzcd} Y \arrow[r] \arrow[d] & T \arrow[d] \\ X \arrow[r] & \pi_0(X) \end{tikzcd}$$

be a cartesian diagram in the category of topological spaces with T profinite. If moreover X is w-local, then Y is w-local, $Y \to X$ is w-local, and the set of closed points of Y is the inverse image of the set of closed points of X.

Proof. We know that Y is spectral and $T = \pi_0(Y)$ by Lemma 1.26. Assume X is w-local and let $X^c \subset X$ be its closed points. The inverse image $Y^c \subset Y$ of X^c maps bijectively onto T, since $X^c \to \pi_0(X)$ is a bijection by Lemma 1.72. (Note that we do not yet know whether Y^c is the set of closed points of Y.) Moreover, Y^c is quasi-compact as a closed subset of the spectral space Y. Hence $Y^c \to \pi_0(Y) = T$ is a homeomorphism by `isHomeomorph_iff_continuous_bijective` ([Sta18, Tag 08YE]). It follows that all points of Y^c are closed in Y: if some $y \in Y^c$ specialized to a distinct closed point y' (which also falls in Y since Y is closed), then both would map to the same point in T, contradicting the bijectivity of $Y^c \to T$.

Conversely, if $y \in Y$ is a closed point, then it is closed in its fibre over T, and its image $x \in X$ is closed in the corresponding (homeomorphic) fibre of $X \to \pi_0(X)$. Since a point in a w-local space is closed if and only if it is closed in its connected component (which is closed), we have $x \in X^c$, so $y \in Y^c$. Thus Y^c is exactly the set of closed points of Y. Moreover, since $Y^c \cong T$, each connected component contains a unique closed point, and for each $y \in Y^c$ the set of generalizations of y is precisely the fibre of $Y \to \pi_0(Y)$ (as any generalization of y lies in the same connected component with y). The lemma follows. □

1.6 w-local rings

Definition 1.74 (w-local rings, [BS15, Definition 2.2.1(i)]). A ring A is *w-local* if Spec(A) is w-local.

Definition 1.75 (w-local ring maps, [BS15, Definition 2.2.1(iii)]). A ring map $f : A \to B$ between w-local rings is *w-local* if the induced map of w-local spaces $\mathrm{Spec}(f) : \mathrm{Spec}(B) \to \mathrm{Spec}(A)$ is w-local.

Proof. Omitted. □

Lemma 1.76. *If $A \to B$ is a surjective ring map and A is w-local, then B is w-local.*

Proof. This is Lemma 1.63. □

Lemma 1.77. *Let A be a w-local ring. Then* $\mathrm{Spec}(A)$ *is set theoretically the disjoint union of the spectra of the local rings at closed points.*

Proof. This follows from the fact that every point specializes to a unique closed point and the spectrum of the local ring at a point is the set of all points specialize to it. □

Lemma 1.78. *Let $A \to B$ be a ring map such that*

1. *$A \to B$ identifies local rings, and*
2. $\mathrm{Spec}(B) \to \mathrm{Spec}(A)$ *is bijective.*

Then $A \to B$ is an isomorphism.

Proof. Since $A \to B$ is flat (can be checked on stalks of B), it has going down. Thus Lemma 1.27 shows for any prime $\mathfrak{q} \subset B$ lying over $\mathfrak{p} \subset A$ we have $B_\mathfrak{q} = B_{\mathfrak{p}B}$. Since $B_\mathfrak{q} = A_\mathfrak{p}$ by assumption, we see that $A_\mathfrak{p} = B_{\mathfrak{p}B}$ for all primes $\mathfrak{p}$ of A. Thus $A = B$ since isomorphism can be checked locally on A. □

Lemma 1.79 ([Sta18, Tag 097E]). *Let $A \to B$ be a w-local ring map between w-local rings such that*

1. *$A \to B$ identifies local rings, and*
2. *$\pi_0(\mathrm{Spec}(B)) \to \pi_0(\mathrm{Spec}(A))$ is bijective.*

Then $A \to B$ is an isomorphism.

Proof. By Lemma 1.78, it suffices to show that $Y = \mathrm{Spec}(B) \to X = \mathrm{Spec}(A)$ is bijective. Let $X^c \subset X$ and $Y^c \subset Y$ be the sets of closed points. By assumption Y^c maps into X^c and the induced map $Y^c \to X^c$ is a bijection since Lemma 1.72 holds. By Lemma 1.77, we see that both X and Y , as sets, are disjoint unions of the spectra of the local rings at closed points. As $A \to B$ identifies local rings, $Y \to X$ is a bijection. □

The goal of the remaining part of this section is to show that for every ring A, there exists a faithfully flat, ind-Zariski A-algebra A_w with A_w w-local. Let A be a ring.

Definition 1.80. Let $f \in A$ be an element and $I \subseteq A$ an ideal. Let $S_{f,I} \subseteq A$ be the multiplicative subset of elements which map to invertible elements of $(A/I)_f$. We define $A_{\widetilde{(I,f)}}$ to be the A-algebra $S_{f,I}^{-1}$ and $\widetilde{(I,f)} \subseteq A_{\widetilde{(I,f)}}$ to be the kernel of the induced map $A_{\widetilde{(I,f)}} \to (A/I)_f$.

If $Z \subseteq \mathrm{Spec}(A)$ is a locally closed subscheme of the form $D(f) \cap V(I)$, we also write $A_{\widetilde{Z}}$ instead of $A_{\widetilde{(I,f)}}$. In the following, when we use the notation $A_{\widetilde{Z}}$ there is always a canonical choice of I and f for Z. In general this notation is justified by the fact that, up to isomorphism, the ring $A_{\widetilde{Z}}$ does not depend on the choice of f and I [Sta18, Tag 096V].

Lemma 1.81 ([Sta18, Tag 096V]). *Let $f \in A$ be an element and $I \subseteq A$ an ideal.*

1. *The map $\mathrm{Spec}(A_{\widetilde{(I,f)}}) \to \mathrm{Spec}(A)$ induces a homeomorphism onto the generalization of $D(f) \cap V(g)$.*
2. *The closed subscheme $V(\widetilde{(I,f)})$ induces an isomorphism onto $D(f) \cap V(I)$.*
3. *If A' is a ring with an element $f' \in A$ and ideal $I' \subseteq A'$ and $A \to A'$ is a ring map that maps $D(f') \cap V(I')$ into $D(f) \cap V(I)$, then there is a unique A-algebra map $A_{\widetilde{(I,f)}} \to A'_{\widetilde{(I',f')}}$ making the obvious diagram with $A \to A'$ commute.*

In particular, every point in $\mathrm{Spec}(A_{\widetilde{(I,f)}})$ specializes to a point in $V(\widetilde{(I,f)})$.

Proof. Let $S = S_{f,I}$, $Z = D(f) \cap V(I)$ and $J = \widetilde{(I,f)}$.

Note that Z is homeomorphic to $\mathrm{Spec}((A/I)_f)$. Since the map $\mathrm{Spec}(S^{-1}A) \to \mathrm{Spec}(A)$ is a homeomorphism onto its image, to show (1) it suffices to show that the image is the set of points of $\mathrm{Spec}(A)$ specializing to $D(f) \cap V(I)$. Let $\mathfrak{p}$ be a prime of A specializing to a $\mathfrak{q}$ in Z, so $I \subseteq \mathfrak{q}$ and $f \notin \mathfrak{q}$. In particular, the image of every element of $\mathfrak{p}$ in $(A/I)_f$ is contained in $\mathfrak{q}$ and is therefore not invertible. Conversely, let $\mathfrak{p}$ be a prime of A which does not specialize to a point in Z, hence the image of $\mathfrak{p}$ in $(A/I)_f$ contains the unit element. So there exists $g \in \mathfrak{p}$ that becomes invertible in $(A/I)_f$, so $g \in S$ and $\mathfrak{p}$ is not in the image of $\mathrm{Spec}(S^{-1}A)$.

By construction, the induced map $S^{-1}A \to (A/I)_f$ is surjective and hence induces an isomorphism $S^{-1}A/J \to (A/I)_f$. This shows (2).

Let $g \in A$. Suppose the image of g in $(A'/I')_{f'}$ is not invertible. Then there exists a prime ideal $\mathfrak{p}$ of $(A'/I')_{f'}$ that contains the image of g. Hence g is in the preimage of $\mathfrak{p}$ under $A \to A'$, which lies in Z by assumption. Thus g is not invertible in $(A/I)_f$, which shows $g \notin S$. □

In view of 1.81, given $Z = D(f) \cap V(I)$ we may view Z as a closed subscheme of $\mathrm{Spec}(A_{\widetilde{Z}})$ defined by the ideal $V(\widetilde{(I, f)})$.

Lemma 1.82. *Let $f \in A$ be an element and $I \subseteq A$ an ideal. Then $A_{\widetilde{(I,f)}}$ is ind-Zariski over A.*

Proof. $A_{\widetilde{(I,f)}} = S^{-1}A$ for some multiplicative subset A. □

Definition 1.83 (Stratification by subsets)**.** Let $E, F \subseteq A$ be subsets. We define

$$Z(E, F) = \bigcap_{f \in E'} D(f) \cap \bigcap_{f \in F} V(f).$$

Lemma 1.84. *Let A be a ring, $E, F \subseteq A$ finite subsets. Then*

$$Z(E, F) = D\left(\prod_{f \in E} f\right) \cap V\left(\sum_{f \in F} fA\right).$$

Proof. This is immediate from the definitions of D and V. □

Lemma 1.85. *Let $E_1, F_1, E_2, F_2 \subseteq A$ be subsets such that $E_1 \subseteq E_2$ and $F_1 \subseteq F_2$. Then $Z(E_2, F_2) \subseteq Z(E_1, F_1)$.*

Proof. This is immediate from the definition of $Z(-,-)$. □

Lemma 1.86. *Let A be a ring, $E \subseteq A$ a finite subset. Then*

$$\mathrm{Spec}(A) = \coprod_{E = E' \coprod E''} Z(E', E'')$$

set theoretically.

Proof. Let $x \in \mathrm{Spec}(A)$ and $E = E' \coprod E''$ a disjoint union decomposition. Then $x \in Z(E', E'')$ if and only if $f(x) \neq 0$ for all $f \in E'$ and $f(x) = 0$ for all $f \in E''$. Hence x is contained in

$$Z(\{f \in E \mid f(x) \neq 0\}, \{f \in E \mid f(x) = 0\}),$$

and this is the only set of this form in which x is contained. □

Definition 1.87. Let A be a ring and $E \subseteq A$ a finite subset. Let

$$A_E = \prod_{E = E' \coprod E''} A_{\widetilde{Z(E', E'')}}.$$

Further, we set $I_E \subseteq A$ to be the product of the ideals defining $Z(E', E'')$ in $\mathrm{Spec}(A_{\widetilde{Z(E', E'')}})$.

Note that 1.87 is well-defined, because for finite subsets E', E'' of A, the locally closed subset $Z(E', E'')$ is always of the right form by 1.84.

Lemma 1.88. *Let $E \subseteq A$ be a finite subset. Then $V(I_E) = \coprod_{E = E' \coprod E''} Z(E', E'')$. In particular, the map $\mathrm{Spec}(A_E) \to \mathrm{Spec}(A)$ induces a bijection $V(I_E) \to \mathrm{Spec}(A)$.*

Proof. By definition I_E is the product of the ideals defining $Z(E',E'')$ in $\mathrm{Spec}(A_{\widetilde{Z(E',E'')}})$. Hence the first claim. The second follows from the first, because $\mathrm{Spec}(A) = \coprod_{E=E'\coprod E''} Z(E',E'')$ by 1.86. $\square$

Lemma 1.89. *Let $E \subseteq A$ be a finite subset. Then every point of $\mathrm{Spec}(A_E)$ specializes to a point in $V(I_E)$. In particular, $V(I_E)$ contains all closed points of $\mathrm{Spec}(A_E)$.*

Proof. Let $x \in \mathrm{Spec}(A_E)$. Then x is in $\mathrm{Spec}(A_{\widetilde{Z(E',E'')}})$ for some $E = E' \coprod E''$. By 1.81, x specializes to a point in $Z(E',E'') \subseteq V(I_E)$. If x is closed, then x specializes to itself, so $x \in V(I_E)$. $\square$

Let A be a ring. Given finite subsets $E_1, E_2 \subseteq A$ with $E_1 \subseteq E_2$, there is a canonical transition map $A_{E_1} \to A_{E_2}$ of A-algebras: For a disjoint union decomposition $E_2 = E_2' \coprod E_2''$, we set $E_1' = E_1 \cap E_2'$ and $E_1'' = E_1 \cap E_2''$. By 1.85, we obtain $Z(E_2',E_2'') \subseteq Z(E_1',E_1'')$ and hence by 1.81 a unique A-algebra map $A_{\widetilde{Z(E_1',E_1'')}} \to A_{\widetilde{Z(E_2',E_2'')}}$.

Lemma 1.90. *Let $E_1, E_2 \subseteq A$ be finite subsets with $E_1 \subseteq E_2$. The induced map $\mathrm{Spec}(A_{E_2}) \to \mathrm{Spec}(A_{E_1})$ maps $V(I_{E_2})$ into $V(I_{E_1})$.*

Proof. This holds because for every decomposition $E_2 = E_2' \coprod E_2''$ and $E_1' = E_1 \cap E_2'$ and $E_1'' = E_1 \cap E_2''$, the induced map $\mathrm{Spec}(A_{\widetilde{Z(E_2',E_2'')}}) \to \mathrm{Spec}(A_{\widetilde{Z(E_1',E_1'')}})$ maps $Z(E_2',E_2'')$ into $Z(E_1',E_1'')$. $\square$

Definition 1.91. Let $I(A)$ be the partially ordered set of all finite subsets of A. We define the A-algebra A_w as

$$A_w = \mathrm{colim}_{E\in I(A)} A_E.$$

We denote by $I_w \subseteq A_w$ the colimit of the ideals I_E, i.e., the union of the images of the I_E in A_w.

By the following lemma, we see $V(I_w)$ is the inverse limit of the closed subsets $V(I_E)$.

Lemma 1.92. *Let $(A_i, I_i)_{i\in I}$ be a filtered system of rings with ideals. Set $A = \mathrm{colim}_{i\in I} A_i$ and $I = \mathrm{colim}_{i\in I} I_i = \mathrm{colim}_{i\in I} I_i A$. Then*

$$V(I) = \lim_{i\in I} V(I_i) = \bigcap_{i\in I} \textit{preimage of } V(I_i) \textit{ in } \mathrm{Spec}(A).$$

Proof. Omitted. $\square$

Lemma 1.93. *The A-algebra A_w is ind-Zariski.*

Proof. For $E \subseteq A$ a finite subset, the A-algebra A_E is ind-Zariski as a finite product of ind-Zariski algebras by 1.43. Thus A_w is a filtered colimit of ind-Zariski algebras, hence ind-Zariski by 1.42. $\square$

Lemma 1.94. *Let A be a ring.*

1. *The composition $V(I_w) \to \mathrm{Spec}(A_w) \to \mathrm{Spec}(A)$ is a bijection.*
2. *Every point of $\mathrm{Spec}(A_w)$ specializes to a unique point of $V(I_w)$.*
3. *$V(I_w)$ is the set of closed points of $\mathrm{Spec}(A_w)$.*

Proof. Since $V(I_w) = \lim_E V(I_E)$ (by Lemma 1.92) and $V(I_E) \to \mathrm{Spec}(A_E) \to \mathrm{Spec}(A)$ is bijective for all E by Lemma 1.88, the first claim holds.

We first show that every point of $\mathrm{Spec}(A_w)$ specializes to a point in $V(I_w)$. For this let $y \in \mathrm{Spec}(A_w)$. For all $E \subseteq A$ finite, denote by T_E the closure of the image of y in $\mathrm{Spec}(A_E)$. By Lemma 1.89, the intersection $T_E \cap V(I_E)$ is non-empty. Since both T_E and $V(I_E)$ are closed, the intersection is closed and hence quasi-compact. Thus $(\lim_E T_E) \cap V(I_w) = \lim_E (T_E \cap V(I_E))$ is non-empty. Apply Lemmas 1.8 and 1.9 to singleton y, the subset $\lim_E T_E$ is the closure of y in $\mathrm{Spec}(A_w)$. (Shuold be a cofiltered version of `IsCompact.nonempty_iInter_of_directed_nonempty_isCompact_isCl`
Hence y specializes to a point in $V(I_E)$.

For uniqueness, suppose $y \in \mathrm{Spec}(A_w)$ specializes to $z, z' \in V(I_w)$ with $z \neq z'$. Denote the images of z, z' in $\mathrm{Spec}(A)$ by x, x'. Because $V(I_w) \to \mathrm{Spec}(A_w) \to \mathrm{Spec}(A)$ is injective, x and x' are distinct. Hence we may assume there exists $f \in A$ such that $x \in D(f)$ and $x' \in V(f)$. Set $E = \{f\}$. Then

$$\mathrm{Spec}(A_E) = \mathrm{Spec}(A_{\widetilde{(0,f)}}) \coprod \mathrm{Spec}(A_{\widetilde{((f),1)}}).$$

Denote the images of y, z, z' in $\mathrm{Spec}(A_E)$ by y_E, x_E, x'_E. Since x_E maps to $x \in D(f)$ and x'_E maps to $x' \in V(f)$, they lie in different components of the disjoint union decomposition above. But since $\mathrm{Spec}(A_w) \to \mathrm{Spec}(A)$ is continuous, y_E specializes to both x_E and x'_E which is not possible.

Finally, every closed point specializes to itself and is hence contained in $V(I_w)$. Conversely, every point in $V(I_w)$ specializes only to itself, so it is closed. □

Corollary 1.95. *Let A be a ring. The A-algebra A_w is w-local.*

Proof. Immediate from Items 2 and 3 in Lemma 1.94. □

Lemma 1.96. *The A-algebra A_w is faithfully flat.*

Proof. By 1.93, $A \to A_w$ is flat. Because $V(I_w)$ surjects onto $\mathrm{Spec}(A)$ by 1.94, in particular $\mathrm{Spec}(A_w) \to \mathrm{Spec}(A)$ is surjective, hence $A \to A_w$ is faithfully flat. □

Lemma 1.97 ([Sta18, Tag 00GA])**.** *Let B be an A-algebra and $\mathfrak{m}$ a maximal ideal of of R. Let $\mathfrak{q}$ be a prime ideal lying over $\mathfrak{m}$ such that $\kappa(\mathfrak{q})$ is algebraic over $\kappa(\mathfrak{m})$. Then $\mathfrak{q}$ is maximal.*

Proof. We have a factorization $\kappa(\mathfrak{m}) = R/\mathfrak{m} \to S/\mathfrak{q} \to \kappa(\mathfrak{q})$. Hence $S/\mathfrak{q}$ is an intermediate subring of an algebraic field extension and therefore a field. □

Lemma 1.98. *Let A be w-local and $I \subseteq A$ an ideal. Then $A_{\widetilde{V(I)}} = A_{\widetilde{(I,1)}}$ is w-local. If $V(I)$ only contains closed points, then the set of closed points is $V(IA_{\widetilde{V(I)}})$.*

Proof. By 1.81, $\mathrm{Spec}(A_{\widetilde{V(I)}})$ is homeomorphic to the generalization $V(I)^g$. Since $\mathrm{Spec}(A)$ is w-local and $V(I)$ is closed under specialization, also $V(I)^g$ is closed under specialization by 1.61. Hence $\mathrm{Spec}(A_{\widetilde{V(I)}}) \cong V(I)^g$ is w-local by 1.62.

By Item 2 in Lemma 1.81, we may identify $V(IA_{\widetilde{V(I)}})$ with $V(I)$. Since every point in $\mathrm{Spec}(A_{\widetilde{V(I)}}) \cong V(I)^g$ specializes to a unique point in $V(I)$, the closed subset $V(I)$ contains all closed points of $\mathrm{Spec}(A_{\widetilde{V(I)}})$. Hence, if $V(I)$ only contains closed points, the closed points are exactly $V(I)$. □

Proposition 1.99 (Universal property of w-localization, [Sta18, Tag 0977])**.** *Let $A \to B$ be a ring map with B w-local. There exists a unique factorization $A \to A_w \to B$ such that $A_w \to B$ is w-local.*

Proof. TBA. ☐

Definition 1.100 (w-localization of ring map)**.** Let $f\colon A \to B$ be a ring map. We denote by $f_w\colon A_w \to B_w$ the unique w-local ring map making

$$\begin{array}{ccc} A & \longrightarrow & B \\ \downarrow & & \downarrow \\ A_w & \longrightarrow & B_w \end{array}$$

commute.

Lemma 1.101. *Let $A \to B$ be a ring map that induces algebraic extensions on residue fields and let $I \subseteq A$ be an ideal, such that $V(I)$ only contains closed points. Then $V(IB)$ only contains closed points.*

Proof. By 1.97, every prime of B lying above a maximal ideal of A is maximal. In particular, the closed subset $V(IB) = (\operatorname{Spec}(B) \to \operatorname{Spec}(A))^{-1}(V(I))$ only contains closed points. ☐

Lemma 1.102. *Let B be an A-algebra satisfying going down. If every closed point of* $\operatorname{Spec}(A)$ *has a preimage in* $\operatorname{Spec}(B)$*, the map* $\operatorname{Spec}(B) \to \operatorname{Spec}(A)$ *is surjective.*

Proof. Let $\mathfrak{p}$ be a prime of A. Then it is contained in a maximal ideal $\mathfrak{m}$ that is the preimage of a prime $\mathfrak{q}$ of B. By going-down, there exists therefore a prime $\mathfrak{q}' \subseteq \mathfrak{q}$ lying above $\mathfrak{p}$. ☐

Definition 1.103. Let A be a ring and $I \subseteq A$ an ideal. We define the *w-localization of A with respect to I* to be the ring $(A_w)_{\widetilde{V(IA_w)}}$, denoted by $A_{w,I}$.

Lemma 1.104. *Let A be a ring and $I \subseteq A$ an ideal. The ring $A_{w,I}$ is w-local.*

Proof. Follows immediately from Lemma 1.98 and Corollary 1.95. ☐

Lemma 1.105. *Let A be a ring and $I \subseteq A$ an ideal. The ring map $A \to A_{w,I}$ is ind-Zariski.*

Proof. The map $A \to A_w$ is ind-Zariski by Lemma 1.93 and the map $A_w \to A_{w,I}$ is ind-Zariski by Lemma 1.82. Hence the composition $A \to A_{w,I}$ is ind-Zariski by Lemma 1.44. ☐

Lemma 1.106. *Let $I \subseteq A$ be an ideal such that $V(I) \subseteq A$ only contains closed points. Then*

1. *the set of closed points of $A_{w,I}$ is $V(IA_{w,I})$, and*
2. *the map $A/I \to A_{w,I}/IA_{w,I}$ is an isomorphism.*

Proof. $A \to A_w$ is ind-Zariski by 1.93 and therefore induces algebraic extensions of residue fields by 1.40. In particular, the closed subset $V(IA_w)$ only contains closed points by 1.101 and the first claim follow from 1.98.

The map $A \to A_{w,I}$ is ind-Zariski as a composition of ind-Zariski maps by 1.93 and 1.82. Since ind-Zariski is stable under base change, the same holds for $A/I \to A_{w,I}/IA_{w,I}$. Hence it identifies local rings. By Item 2 in Lemma 1.81, $V(IA_{w,I}) \to V(IA_w)$ is an isomorphism. Moreover $V(IA_w) \to V(I)$ is a bijection by Item 1 because $V(IA_w) \subseteq V(I_w)$. Hence, the composition $\operatorname{Spec}(A_{w,I}/IA_{w,I}) = V(IA_{w,I}) \to V(IA_w) \to V(I)$ is bijective. Thus $A/I \to A_{w,I}/IA_{w,I}$ is an isomorphism by 1.78. ☐

Lemma 1.107 ([Sta18, Tag 097A], (2))**.** *Let A be a w-local ring. Let $I \subset A$ be an ideal cutting out the set X^c of closed points in $X = \operatorname{Spec}(A)$. Let $A \to B$ be a ring map inducing algebraic extensions on residue fields at primes. Let C be the ring $B_{w,IB}$ constructed in Definition 1.103. Then*

1. *$B/IB \to C/IC$ is an isomorphism,*
2. *C is w-local,*
3. *the map $p : Y = \operatorname{Spec}(C) \to X$ induced by the ring map $A \to B \to C$ is w-local, and*
4. *$p^{-1}(X^c)$ is the set of closed points of Y, i.e. $V(IC)$ is the set of closed points of Y.*

Proof. Every point of $V(IB)$ is a closed point of $\operatorname{Spec}(B)$ by Lemma 1.101. Hence we may choose $C = (B_w)_{V(\widetilde{IB_w})}$ as in Definition 1.103. Item 1 holds by Lemma 1.106. Item 2 holds by Lemma 1.104. Then the set of closed points of C is $V(IC)$, which is the preimage of $V(I)$ under $\operatorname{Spec}(C) \to \operatorname{Spec}(B) \to \operatorname{Spec}(A)$. □

Lemma 1.108. *Let A be a w-local ring. Let $I \subset A$ be an ideal cutting out the set X^c of closed points in $X = \operatorname{Spec}(A)$. Let B be a faithfully flat A-algebra. Let C be the ring $B_{w,IB}$ constructed in Definition 1.103. Then the composition $A \to B \to C$ is also faithfully flat.*

Proof. By Lemmas 1.41 and 1.105, $A \to B \to C$ is flat, so by 1.102 it suffices to show every closed point of A is in the image. But since $\operatorname{Spec}(B) \to \operatorname{Spec}(A)$ is surjective, also $V(IC) \cong V(IB) \to V(I)$ is surjective (the first map is an isomorphism by Item 2 in Lemma 1.106). □

1.7 w-strictly-local Rings

Definition 1.109 (w-strictly local rings, [BS15, Definition 2.2.1(ii)])**.** A ring A is *w-strictly local* if A is w-local, and every local ring of A at a maximal ideal is strictly henselian.

The goal of this section is to show that every ring A admits a faithfully flat, ind-étale algebra B that is *w*-strictly-local.

Lemma 1.110. *Let A be a strictly henselian local ring. Let $A \to B$ be an ind-étale ring map. Let $\mathfrak{n}$ be an maximal ideal of B lying over the maximal ideal $\mathfrak{m}$ of A. Then the map $A \to B_{\mathfrak{n}}$ is isomorphism.*

Proof. Write $B = \operatorname{colim}_i B_i$ as a filtered colimit of étale A-algebras B_i. Since localization commutes with colimits, we have $B_{\mathfrak{n}} = \operatorname{colim}_i (B_i)_{\mathfrak{n}_i}$ where $\mathfrak{n}_i$ is the restriction of $\mathfrak{n}$ to B_i. Then use Proposition 1.51 to conclude each B_i is isomorphic to A. □

Lemma 1.111. *Let A be a ring such that every faithfully flat étale ring map $A \to B$ has a retraction. Then every local ring of A at a maximal ideal is strictly henselian.*

Proof. Let $\mathfrak{m}$ be a maximal ideal of A, denote by κ the residue field $\kappa(\mathfrak{m})$ and by κ^{sep} a separable closure of κ. Let $A_{\mathfrak{m}} \to B \to \kappa^{\text{sep}}$ be a factorisation of $A_{\mathfrak{m}} \to \kappa^{\text{sep}}$ where $A_{\mathfrak{m}} \to B$ is étale. Since $A_{\mathfrak{m}} = \operatorname{colim}_{f \notin \mathfrak{m}} A_f$, by 1.37 there exists $f \notin \mathfrak{m}$ and an étale A_f-algebra B' such that $B = A_{\mathfrak{m}} \otimes_{A_f} B'$. Because $A \to A_f \to B'$ is étale, there are finitely many prime ideals $\mathfrak{q}_1, \dots, \mathfrak{q}_n$ of B' lying over $\mathfrak{m}$. Since the kernel of the composition $B' \to B \to \kappa^{\text{sep}}$ lies over $\mathfrak{m}$ (the kernel of $A \to \kappa_{\text{sep}}$ is $\mathfrak{m}$), we have $n \geq 1$. Set $\mathfrak{q} = \mathfrak{q}_1$. By prime avoidance, there exists $g \in B'$ such that $g \in \mathfrak{q}_i$ for all $i \geq 2$ and $g \notin \mathfrak{q}$. Hence $\mathfrak{q}$ is the unique prime ideal of B'_g lying over $\mathfrak{m}$.

Let U be the image of the induced map $\mathrm{Spec}(B'_g) \to \mathrm{Spec}(A)$. Since $A \to B'$ is étale, U is open. Since $\{\mathfrak{m}\}$ is closed in $\mathrm{Spec}(A)$, there exist a_i such that

$$\mathrm{Spec}(A) - \{\mathfrak{m}\} = \bigcup_i D(a_i).$$

The complement of U is closed and quasicompact, and contained in $\mathrm{Spec}(A) - \{\mathfrak{m}\}$, hence there exist $a_1, \ldots, a_r$ such that

$$\mathrm{Spec}(A) = U \cup \bigcup_{i=1}^{r} D(a_i).$$

Hence the map $A \to B'_g \times \prod_{i=1}^r A_{a_i}$ is étale and faithfully flat. By assumption, it therefore has a retraction σ. Since $\mathfrak{q}$ is the unique prime ideal of $B'_g \times \prod_{i=1}^r A_{a_i}$ lying over $\mathfrak{m}$, we have $\sigma^{-1}(\mathfrak{m}) = \mathfrak{q}$. In particular, we obtain a retraction

$$B'_{\mathfrak{q}} = \left(B'_g \times \prod_{i=1}^{r} A_{a_i} \right)_{\mathfrak{q}} \to A_{\mathfrak{m}}$$

of the map $A_{\mathfrak{m}} \to B'_{\mathfrak{q}}$. Precomposing with $B' \to B'_{\mathfrak{q}}$ we obtain a map $B' \to A_{\mathfrak{m}}$ and hence a map $B = B' \otimes_{A_f} A_{\mathfrak{m}} \to A_{\mathfrak{m}}$ that is a retraction of $A_{\mathfrak{m}} \to B$. □

Definition 1.112. Let A be a ring. Denote by $S(A)$ the set of finite subsets of faithfully flat étale A-algebras. We call the A-algebra $T(A) = \mathrm{colim}_{E \in S(A)} \bigotimes_{B \in E} B$ the *étale precontraction* of A.

Lemma 1.113. *For every faithfully flat étale ring map $A \to B$, there exists an A-algebra map $B \to T(A)$.*

Proof. $\{B\}$ is an element of $S(A)$, hence B itself is part of the diagram defining $T(A)$. □

Lemma 1.114. *Let R be a commutative ring.*

- *The forgetful functors from commutative R-algebras to R-algebras and from R-algebras to R-modules preserve filtered colimits.*
- *The forgetful functor from commutative R-algebras to commutative rings preserves filtered colimits.*
- *Tensoring on the left with an R-algebra preserves colimits on the category of commutative R-algebras.*

Lemma 1.115. *Let $B = \mathrm{colim}_i B_i$ be a filtered colimit of (faithfully) flat A-algebras. Then B is (faithfully) flat over A.*

Proof. Because colimits commute with tensor products, ind-flat is flat. One also checks that pro-surjective is surjective, hence the claim. □

Lemma 1.116. *$T(A)$ is ind-étale and faithfully flat over A.*

Proof. For every $E \in S(A)$, the A-algebra $\bigotimes_{B \in E} B$ is étale and faithfully flat. Hence the result follows from the definition of ind-étale and Lemma 1.115. □

Definition 1.117. Let A be a ring. Set $T^0(A) = A$ and for $n \in \mathbb{N}$ set $T^{n+1}(A) = T(T^n(A))$. We call the A-algebra $T^\infty(A) = \operatorname{colim}_{n\in\mathbb{N}} T^n(A)$ the *étale contraction of A*.

Lemma 1.118. *$T^\infty(A)$ is ind-étale and faithfully flat over A.*

Proof. This follows from Lemma 1.116, Lemma 1.115 and Lemma 1.55. □

Proposition 1.119 ([Sta18, Tag 097R])**.** *Let $T^\infty(A) \to B$ be a faithfully flat and étale A-algebra map. Then it has a retraction.*

Proof. By Lemma 1.37 there exists $n \in \mathbb{N}$ and a faithfully flat, étale $T^n(A)$-algebra B' such that $B = T^\infty(A) \otimes_{T^n(A)} B'$. By Lemma 1.113, there exists a map $B' \to T^{n+1}(A)$. The identity of $T^\infty(A)$ and the composition $B' \to T^{n+1}(A) \to T^\infty(A)$ now induce a retraction

$$B = T^\infty(A) \otimes_{T^n(A)} B' \to T^\infty(A).$$

□

Corollary 1.120. *Let $\mathfrak{m}$ be a maximal ideal of $T^\infty(A)$. Then $T^\infty(A)_\mathfrak{m}$ is strictly Henselian.*

Proof. This follows from Proposition 1.119 and Lemma 1.111. □

Lemma 1.121. *Let k be a field and A an ind-étale, local k-algebra. Then A is a separable algebraic field extension of k.*

Proof. Let $A = \operatorname{colim}_i A_i$ such that A_i is an étale k-algebra for all i. Let $\mathfrak{m}$ be the maximal ideal of A. Since localization commutes with colimits, we obtain $A = A_\mathfrak{m} = \operatorname{colim}_i (A_i)_{\mathfrak{m}_i}$ where $\mathfrak{m}_i$ is the restriction of $\mathfrak{m}$ to A_i. Since A_i is étale over k, it is a finite product of finite separable extensions of k, hence $(A_i)_{\mathfrak{m}_i}$ is a finite separable extension of k. Since a filtered colimit of finite separable extensions is an algebraic separable extension, we conclude. □

Proposition 1.122. *Let $f \colon A \to B$ be ind-étale. Then f induces separable algebraic extensions on residue fields.*

Proof. Let $\mathfrak{q}$ be a prime ideal of B lying over $\mathfrak{p} \subseteq A$. Write $B = \operatorname{colim}_i B_i$ as a filtered colimit of étale A-algebras. Every element $s \in B$ lifts to some $s_i \in B_i$, and the composition

$$\kappa(\mathfrak{p}) \otimes_A B_i \to \kappa(\mathfrak{q})$$

is an algebra homomorphism from the étale $\kappa(\mathfrak{p})$-algebra $\kappa(\mathfrak{p}) \otimes_A B_i$ to the local ring $\kappa(\mathfrak{q})$, so by 1.121 the image of s in $\kappa(\mathfrak{q})$ is separable over $\kappa(\mathfrak{p})$. A general element of $\kappa(\mathfrak{q})$ is a ratio of two such images, and separability passes to ratios via the separable closure. □

Proposition 1.123 ([Sta18, Tag 097X])**.** *For any ring A, there exists an ind-étale faithfully flat A-algebra B with B w-strictly local.*

Proof. By Lemma 1.94, the ring A_w is w-local, so the set of closed points of $\operatorname{Spec}(A_w)$ is closed. Let I be the radical ideal of A_w such that $V(I)$ is the set of closed points of $\operatorname{Spec}(A_w)$. Set $B' = T^\infty(A_w)$. By Corollary 1.120, the local ring $B'_\mathfrak{m}$ is strictly henselian for every maximal ideal $\mathfrak{m}$ of B'. The map $A_w \to B'$ induces algebraic extensions on residue fields at primes by Lemma 1.118 and Proposition 1.122, so applying Definition 1.103 and Lemma 1.107 to $A_w \to B'$, we obtain an ind-Zariski ring map $B' \to B$ such that $B'/IB' \to B/IB$ is an isomorphism, $V(IB)$ is the set of closed points of $\operatorname{Spec}(B)$, B is w-local and $A_w \to B' \to B$ is faithfully flat by Lemma 1.108. Since ind-Zariski maps identify local rings by Lemma 1.40 and every closed point of $\mathrm{Spec}(B)$ maps to a closed point of B', the local rings at maximal ideals of B are strictly Henselian. Hence B is w-strictly local.

It remains to verify that the composition $A \to A_w \to B' \to B$ is ind-étale and faithfully flat. This follows because both ind-étale and faithfully flat is stable under composition. $\square$

Remark 1.124. By following all constructions, we may choose B as $((T^\infty(A_w))_w)_{V(I((\widetilde{T^\infty(A_w)})_w))}$ where I is the radical ideal of A_w defining the closed points of A_w.

One may ask why $T^\infty(A)_w$ does not work. The reason for this is that the map $\mathrm{Spec}(T^\infty(A)_w) \to \mathrm{Spec}(T^\infty(A))$ does not necessarily map closed points to closed points. If A is already w-local, we can eliminate the points mapping to non-closed points by taking I to be the ideal defining the closed points of $\mathrm{Spec}(A)$. Then $V(IT^\infty(A))$ only contains closed points and by passing to the generalization of the preimage of $V(IT^\infty(A))$ in $\mathrm{Spec}(T^\infty(A)_w)$, we obtain the desired affine scheme. If A is not w-local, we may replace A by A_w in this process and obtain the result.

Definition 1.125 ([Sta18, Tag 097C])**.** Let A be a ring, $X = \mathrm{Spec}(A)$, and $T \subset \pi_0(X)$. Define the ring A_T by

$$A_T := \mathrm{colim}_W A_W,$$

where W runs over all open and closed subsets of X containing the inverse image Z of T, and $A \to A_W$ is the localization of A at W.

Lemma 1.126 ([Sta18, Tag 097C])**.** *Let A be a ring, $X = \mathrm{Spec}(A)$, and $T \subset \pi_0(X)$ a closed subset. Then the map $A \to A_T$ from Definition 1.125 is a surjective ind-Zariski ring map, and the induced morphism $\mathrm{Spec}(A_T) \to \mathrm{Spec}(A)$ restricts to a homeomorphism of $\mathrm{Spec}(A_T)$ onto the inverse image of T in X.*

Proof. Each $A \to A_W$ is a localization at an idempotent. Passing to filtered colimits, $A \to A_T$ is surjective and ind-Zariski.

When T is closed, its inverse image Z is a closed union of connected components. By Lemma 1.16, we have

$$Z = \bigcap_{\substack{Z \subseteq W \\ W \text{ open and closed}}} W.$$

Thus, $\mathrm{Spec}(A_T) = \varprojlim_W W$ is homeomorphic to Z via the natural map. $\square$

In the following series of constructions, we will modify the connected components of a spectrum to match a given profinite space as in [Sta18, Tag 097D].

Definition 1.127. Let A be a ring and let $X = \mathrm{Spec}(A)$. Let T be profinite space and let $f : T \to \pi_0(X)$ be a continuous map. Let S be a finite discrete quotient of T with quotient map $q : T \to S$. Define the ring A_q^f as follow: let $Z = \mathrm{Im}(T \to \pi_0(X) \times S = \pi_0(\mathrm{Spec}(A^S)))$, where $A^S = \prod_{s \in S} A$ and let

$$A_q^f := (A^S)_Z$$

be the A^S-algebra as in Definition 1.125.

Lemma 1.128. *Let A be a ring and let $X = \mathrm{Spec}(A)$. Let T be profinite space and let $f : T \to \pi_0(X)$ be a continuous map. Let S be a finite discrete quotient of T with quotient map $q : T \to S$. Then the A^S-algebra defined in Definition 1.127 viewed as A-algebra through $A \to A^S \to A_q^f$ is ind-Zariski. In particular, it identifies local rings. The map $A^S \to A_q^f$ induces a homeomorphism of $\mathrm{Spec}(A_q^f)$ onto $(X \times S) \times_{\pi_0(X) \times S} Z = X \times_{\pi_0(X)} Z$.*

Proof. The map $A \to A^S$ is ind-Zariski by Lemma 1.43 and $A^S \to A^f_q$ is also ind-Zariski by Lemma 1.126. The composition $A \to A^f_q$ is therefore ind-Zariski as well by Lemma 1.44. By Lemma 1.40, $A \to A^f_q$ identifies local rings. The last sentence is a direct consequence of Lemma 1.126. ☐

Definition 1.129. Let A be a ring and let $X = \mathrm{Spec}(A)$. Let T be profinite space and let $f : T \to \pi_0(X)$ be a continuous map. Let S and S' be finite discrete spaces and let $q : T \to S$ and $q' : T \to S'$ be quotient maps. Let $g : S' \to S$ be a map such that $q = g \circ q'$. Let $Z \subseteq \pi_0(\mathrm{Spec}(A^S)) \times S$ and $Z' \subseteq \pi_0(\mathrm{Spec}(A^{S'})) \times S'$ be the images of T under the maps induced by f and f' respectively. Then g induces a map $Z \to Z'$ and a map $\tilde{g} : \mathrm{Spec}(A^f_q) \cong X \times_{\pi_0(X)} Z \to X \times_{\pi_0(X)} Z' \cong \mathrm{Spec}(A^{f'}_{q'})$.

Since both maps $A \to A^f_q$ and $A \to A^{f'}_{q'}$ identify local rings, by the bijection in Definition 1.35 we can find a transition ring map

$$t_g : A^f_q \to A^{f'}_{q'}$$

that induces the topological map $\tilde{g}$ between the spectra.

Lemma 1.130. *Let A be a ring and let $X = \mathrm{Spec}(A)$. Let T be profinite space and let $f : T \to \pi_0(X)$ be a continuous map. Let S, S' and S'' be finite discrete spaces and let $q : T \to S$, $q' : T \to S'$ and $q'' : T \to S''$ be quotient maps. Let $g : S' \to S$ and $g' : S'' \to S'$ be maps such that $q = g \circ q'$ and $q' = g' \circ q''$. Then the transition maps defined in Definition 1.129 satisfy*

$$t_{g'} \circ t_g = t_{g \circ g'}.$$

Proof. Both maps $t_{g'}$ and $t_g \circ t_{g' \circ g}$ induce the same topological map between the spectra by construction. Since all maps involved identify local rings by Lemma 1.128, by fully faithfulness in Definition 1.35 we conclude that they are equal. ☐

Definition 1.131 ([Sta18, Tag 097D])**.** Let A be a ring and let $X = \mathrm{Spec}(A)$. Let T be a profinite space and let $T \to \pi_0(X)$ be a continuous map. Write $T = \lim T_i$ as the limit of an inverse system of finite discrete spaces over a directed set. Define the ring $A^f_{\pi_0}$ as follows:

$$A^f_{\pi_0} := \mathrm{colim}_{(T \to T_i)} A^f_{q_i},$$

where $q_i : T \to T_i$ are the quotient maps and $A^f_{q_i}$ are the rings defined in Definition 1.127 and the transition maps are those defined in Definition 1.129. By Lemma 1.130 the transition maps are compatible thus the colimit is well-defined.

Lemma 1.132 ([Sta18, Tag 097D])**.** *Let A be a ring and let $X = \mathrm{Spec}(A)$. Let T be a profinite space and let $T \to \pi_0(X)$ be a continuous map. Then the A-algebra $A^f_{\pi_0}$ defined in Definition 1.131 is ind-Zariski. Let $Y = \mathrm{Spec}(A^f_{\pi_0})$. Then $\pi_0(Y) \cong T$ and the diagram*

$$\begin{array}{ccc} Y & \longrightarrow & T \\ \downarrow & & \downarrow \\ X & \longrightarrow & \pi_0(X) \end{array}$$

is cartesian in the category of topological spaces.

Proof. By Lemma 1.128 we see that each $A \to A^f_{q_i}$ is ind-Zariski, thus $A^f_{\pi_0}$ is ind-Zariski by Lemma 1.42. We have the following canonical cartesian diagram

$$\begin{tikzcd} \mathrm{Spec}(A^f_{q_i}) \arrow[r] \arrow[d] & Z_i \arrow[d] \\ X \arrow[r] & \pi_0(X). \end{tikzcd}$$

Because $T = \lim Z_i$ and, $Y = \mathrm{Spec}(A^f_{\pi_0}) = \lim \mathrm{Spec}(A^f_{q_i})$ fits into the cartesian diagram

$$\begin{tikzcd} Y \arrow[r] \arrow[d] & T \arrow[d] \\ X \arrow[r] & \pi_0(X) \end{tikzcd}$$

of topological spaces. By Lemma 1.26 we conclude that $T = \pi_0(Y)$. ◻

Lemma 1.133 ([Sta18, Tag 09AZ, second part in the proof of (3) $\implies$ (1)])**.** *Let A be a ring. If the following two conditions hold:*

1. *A is w-local, and*
2. *$\pi_0(\mathrm{Spec}(A))$ is extremally disconnected.*

Then every faithfully flat ring map $A \to B$ identifying local rings with B w-local and whose closed points of $\mathrm{Spec}(B)$ are exactly $V(IB)$ has a retraction.

Proof. Let $A \to B$ be faithfully flat and identifying local rings, with B w-local and whose closed points of $\mathrm{Spec}(B)$ are exactly $V(IB)$. We will show that $A \to B$ has a retraction.

Choose a continuous section to the surjective continuous map $V(IB) \to V(I)$. This is possible because $V(I) = \mathrm{Spec}(A)^c \cong \pi_0(\mathrm{Spec}(A))$ (by Lemma 1.72) is extremally disconnected, see Theorem 1.2. The image is a closed subspace $T \subset \pi_0(\mathrm{Spec}(B)) \cong V(IB)$ (being the continuous image of the quasi-compact space $V(I)$) that maps homeomorphically onto $\pi_0(\mathrm{Spec}(A))$.

Let $B \to B_T$ be the ring map from Definition 1.125, which is surjective and ind-Zariski by Lemma 1.126. It also induces a homeomorphism $\pi_0(\mathrm{Spec}(B_T)) \cong T \cong \pi_0(\mathrm{Spec}(A))$. Moreover, by Lemma 1.76, B_T is w-local and $B \to B_T$ is w-local. Since ind-Zariski maps identify local rings (Lemma 1.40), the composition $A \to B \to B_T$ remains w-local (Lemma 1.60) and identifies local rings (Lemma 1.31). Therefore, by Lemma 1.79, $A \to B_T$ is an isomorphism. ◻

1.8 w-contractible rings

Definition 1.134 (w-contractible rings, [BS15, Definition 2.4.1])**.** A ring A is *w-contractible* if it is w-strictly local and $\pi_0(\mathrm{Spec}(A))$ is extremally disconnected.

We will see that if a ring A is w-contractible, every faithfully flat, ind-étale map $A \to B$ has a retraction. To show this we will perform a series of reductions.

Lemma 1.135. *Let A be w-contractible and $I \subseteq A$ an ideal cutting out the set X^c of closed points in $X = \mathrm{Spec}(A)$. Then every faithfully flat ind-étale map $A \to B$ with B w-local and whose closed points of $\mathrm{Spec}(B)$ are exactly $V(IB)$ has a retraction.*

Proof. Let $A \to B$ be a faithfully flat ind-étale map with B w-local and such that the closed points of $\mathrm{Spec}(B)$ are exactly $V(IB)$. In this case, $A \to B$ identifies local rings. To justify this, it suffices to check the condition at the maximal ideals of B, which lie over maximal ideals of A since they map to $V(I) = (\mathrm{Spec}(A))^c$. Since A is w-strictly local, the local rings of A at its maximal ideals are strictly Henselian.

Now fix a maximal ideal $\mathfrak{n}$ of B and let $\mathfrak{m} = \mathfrak{n} \cap A$ be its preimage in A, which is a maximal ideal. By base change to $A_{\mathfrak{m}}$ and applying Lemma 1.54, we reduce to the case where B is ind-étale over a strictly Henselian local ring A, with a maximal ideal $\mathfrak{n}$ lying over the maximal ideal $\mathfrak{m}$ of A. We then want to show $B_{\mathfrak{n}} \cong A$. This follows from Lemma 1.110.

Having established that $A \to B$ identifies local rings, we conclude by Lemma 1.133 that it admits a retraction. □

Proposition 1.136 ([Sta18, Tag 0982, (3) $\implies$ (2)])**.** *If A is w-contractible, every ind-étale, faithfully flat map $A \to B$ has a retraction.*

Proof. With Lemma 1.135, it suffices to reduce to the case that B is furthur w-local and the closed points of $\mathrm{Spec}(B)$ are exactly $V(IB)$.

Let $A \to B$ be faithfully flat and ind-étale. We may replace B by the ring $C = B_{w,IB}$ (see Definition 1.103). To justify this, note that the map $A \to C$ is faithfully flat by Lemma 1.108 and $B \to C$ is ind-Zariski by Lemma 1.105, thus ind-étale by Lemma 1.56. Therefore, by Lemma 1.53, the composition $A \to C$ is also ind-étale. Hence, we may assume $\mathrm{Spec}(B)$ is w-local such that the set of closed points of $\mathrm{Spec}(B)$ is $V(IB)$ by Item 2 and **??** in Lemma 1.107.

□

Lemma 1.137. *If a ring C is w-strictly local, then there exists an ind-Zariski faithfully flat C-algebra D with D w-contractible.*

Proof. Choose an extremally disconnected space T and a surjective continuous map $T \to \pi_0(\mathrm{Spec}(C))$ by Proposition 1.5. (T can be chosen as $\beta((\pi_0(\mathrm{Spec}(C)))^{\mathrm{disc}})$.) Note that T is profinite. By Lemma 1.132, Definition 1.131 gives an ind-Zariski ring map $C \to D$ such that $\pi_0(\mathrm{Spec}(D)) \to \pi_0(\mathrm{Spec}(C))$ realizes $T \to \pi_0(\mathrm{Spec}(C))$ and such that

$$\begin{tikzcd}
\mathrm{Spec}(D) \arrow[r] \arrow[d] & \pi_0(\mathrm{Spec}(D)) \arrow[d] \\
\mathrm{Spec}(C) \arrow[r] & \pi_0(\mathrm{Spec}(C))
\end{tikzcd}$$

is cartesian in the category of topological spaces.

Following Lemma 1.73, we note that $\mathrm{Spec}(D)$ is w-local, that $\mathrm{Spec}(D) \to \mathrm{Spec}(C)$ is w-local, and that the set of closed points of $\mathrm{Spec}(D)$ is the inverse image of the set of closed points of $\mathrm{Spec}(C)$.

Thus it is still true that the local rings of D at its maximal ideals are strictly henselian (as they are isomorphic to the local rings at the corresponding maximal ideals of C by Lemma 1.40). Hence D is w-contractible. □

Theorem 1.138 ([BS15, Theorem 2.4.9], [Sta18, Tag 0983])**.** *For any ring A, there exists an ind-étale faithfully flat A-algebra D with D w-contractible.*

Proof. This is a combination of Proposition 1.123 and Lemma 1.137. □

Corollary 1.139. *For any ring A, there exists an ind-étale faithfully flat A-algebra B such that every ind-étale faithfully flat map $B \to C$ has a retraction.*

Proof. This follows from Theorem 1.138 and Proposition 1.136. □

Chapter 2

More on local structure

This chapter serves for the second half of the commutative algebra preparation of the remaining part of the paper. The key result is

1. Theorem 2.46 expressing that every weakly etale map can be covered by ind-etale maps.

The proof goes via the local input .

Lemma 2.1 ([Sta18, Tag 04GH, (1)])**.** *Let R be a Henselian local ring and S a finite R-algebra. Then S is a finite product of Henselian local rings, each finite over R.*

Proof. Omitted, will be in mathlib soon. □

Lemma 2.2. *A filtered colimit of local rings along local homomorphisms is local.*

Proof. Let $R = \operatorname{colim}_i R_i$ with R_i local rings and $R_i \to R_j$ a local ring map. Suppose $x, y \in R_i$ are non-units. Since R_i is local, their sum is a not a unit. If the image of $x + y$ in $\operatorname{colim}_i R_i$ is a unit, their exists a j such that the image of $x + y$ is a unit in R_j. But this does not happen because $R_i \to R_j$ is a local ring map. □

Lemma 2.3 ([Sta18, Tag 04GI])**.** *A filtered colimit of (strictly) Henselian local rings along local homomorphisms is (strictly) Henselian.*

Proof. Omitted. □

Lemma 2.4. *Let R be a Henselian local ring and S be an integral R-algebra and an integral domain. Then S is a local ring.*

Proof. Since S is integral over R, it is the colimit of its finite R-sub-algebras. Since a subring of an integral domain is still an integral domain, by Lemma 2.2 we may assume that S is finite over R. By Lemma 2.1, S is a product of (Henselian) local rings and hence local as an integral domain. □

Lemma 2.5. *Let $R \to S$ be an integral ring map of local rings. Then $R \to S$ is a local ring homomorphism.*

Proof. This follows from going-up. □

Lemma 2.6. *Let $R \to S$ be a local ring homomorphism of local rings. If $R \to S$ is integral, the extension of residue fields $\kappa(S)/\kappa(R)$ is algebraic.*

Proof. It suffices to show that $\kappa(S)$ is integral over R. Since S is integral over R and $\kappa(S)$ is integral over S (the map $S \to \kappa(S)$ is surjective), the composition $R \to S \to \kappa(S)$ is integral. $\square$

Lemma 2.7 ([Sta18, Tag 092Y])**.** *Let $R \to S$ and $R \to T$ be local ring maps of local rings. Suppose $R \to S$ is integral and $\kappa(S)/\kappa(R)$ or $\kappa(T)/\kappa(S)$ is purely inseparable, then $T \otimes_R S$ is a local ring.*

Proof. Integral is stable under base change, so $T \to S' = T \otimes_R S$ is integral. Hence by going-up any maximal ideal of S' lies over $\mathfrak{m}_T$. We have

$$S'/\mathfrak{m}_T S' = \kappa(T) \otimes_R S = \kappa(T) \otimes_{\kappa(R)} S/\mathfrak{m}_R S.$$

Again by going-up and since S is local, $\mathrm{Spec}(S/\mathfrak{m}_R S)$ is a single point. Thus $\mathrm{Spec}(S'/\mathfrak{m}_T S') \cong \mathrm{Spec}(\kappa(T) \otimes_{\kappa(R)} \kappa(S))$. The latter is a single point, because purely inseparable extensions induce universal homeomorphisms. $\square$

2.1 Weakly étale algebras

Ind-étale is a practical notion, since it usually allows reducing proofs to the étale case. Geometrically, ind-étale is badly behaved though, since it is not local on the (geometric) target. For this reason, when we later define the pro-étale site of a scheme, we use the notion of weakly étale ring maps instead.

Definition 2.8 (Weakly étale algebras)**.** An R-algebra S is *weakly étale* if it is flat over R and flat over $S \otimes_R S$. A ring homorphism $f\colon R \to S$ is *weakly étale* if S is weakly étale as an R-algebra.

Historically, the property weakly étale is studied under the name of *absolutely flat*. Lemma 2.15 partially justifies this name. Furthermore, we have an absolute version of this property.

Definition 2.9. A ring A is called *absolutely flat* if every A-module is flat over A.

Lemma 2.10. *Every étale algebra is weakly étale.*

Proof. Let S be an étale R-algebra. Then S is R-flat. Also $S \otimes_R S \cong S \times T$ for some ring T, in particular $\mathrm{Spec}(S \otimes_R S) \to \mathrm{Spec}(S)$ is an open immersion, hence flat. $\square$

Lemma 2.11. *A field is absolutely flat.*

Proof. Omitted. $\square$

Lemma 2.12 ([Sta18, Tag 092J])**.** *If $A \to B$ and $B \to C$ are weakly étale, then $A \to C$ is weakly étale.*

Proof. Omitted, see [Sta18, Tag 092J]. $\square$

Lemma 2.13 ([Sta18, Tag 092J])**.** *For any commutative ring A and multiplicative submonoid $S \subseteq A$, the localization map $A \to S^{-1}A$ is weakly étale.*

Proof. The localization $A \to S^{-1}A$ is flat. Moreover, the multiplication map $S^{-1}A \otimes_A S^{-1}A \to S^{-1}A$ is bijective, hence flat. $\square$

Lemma 2.14 ([Sta18, Tag 092H])**.** *If B is a weakly-étale A-algebra and A' is any A-algebra, the base change $A' \otimes_A B$ is a weakly-étale A'-algebra.*

Proof. Calculation, see [Sta18, Tag 092H]. □

Lemma 2.15 ([Sta18, Tag 092C])**.** *Let B be an A-algebra such that $B \otimes_A B \to B$ is flat. Let N be a B-module. If N is flat as a A-module, then N is flat as a B-module.*

Proof. First proof:

If N' is a second B-module, then

$$N \otimes_B N' = B \otimes_{B \otimes_A B} (N \otimes_A N').$$

Hence this follows from the definitions.

Second proof by element chasing (which is the proof we actually formalized):

To prove that M is flat as a B-module, we will use the equational criterion for flatness: we must show that for any finite relation $\sum_{i=1}^n b_i m_i = 0$ in M (where $b_i \in B$ and $m_i \in M$), there exist elements $m'_s \in M$ and $d_{is} \in B$ such that $m_i = \sum_{s=1}^S d_{is} m'_s$ for all i, and $\sum_{i=1}^n b_i d_{is} = 0$ for all s.

Step 1: The flat epimorphism property.
Let $I = \ker(\mu)$. Since B is a flat $B \otimes_A B$-module, I is a pure ideal. Consider the elements $z_i = 1 \otimes b_i - b_i \otimes 1 \in B \otimes_A B$. Clearly, $\mu(z_i) = b_i - b_i = 0$, so $z_i \in I$ for all $i = 1, \dots, n$.

By the pureness of I, there exists an element $t = \sum_{k=1}^K u_k \otimes v_k \in B \otimes_A B$ such that $\mu(t) = 1$ and $tz_i = 0$ for all i. The condition $\mu(t) = 1$ means $\sum_{k=1}^K u_k v_k = 1$. The condition $tz_i = 0$ implies $t(1 \otimes b_i) = t(b_i \otimes 1)$, which expands to:

$$\sum_{k=1}^K u_k \otimes v_k b_i = \sum_{k=1}^K b_i u_k \otimes v_k \quad \text{in } B \otimes_A B. \tag{2.1}$$

Step 2: Constructing a relation in $B \otimes_A M$.
Define the elements $y_i \in B \otimes_A M$ for $i = 1, \dots, n$ by:

$$y_i = \sum_{k=1}^K u_k \otimes (v_k m_i).$$

Let us evaluate the sum $\sum_{i=1}^n b_i y_i$ in the B-module $B \otimes_A M$ (where B acts on the left factor):

$$\sum_{i=1}^n b_i y_i = \sum_{i=1}^n \sum_{k=1}^K (b_i u_k) \otimes (v_k m_i).$$

Because the map $x \otimes y \mapsto x \otimes y m_i$ from $B \otimes_A B \to B \otimes_A M$ is a well-defined A-linear homomorphism, we can apply it to equation (2.1) to shift the b_i factor:

$$\sum_{k=1}^K (b_i u_k) \otimes (v_k m_i) = \sum_{k=1}^K u_k \otimes (v_k b_i m_i).$$

Summing this over all i yields:

$$\sum_{i=1}^n b_i y_i = \sum_{i=1}^n \sum_{k=1}^K u_k \otimes (v_k b_i m_i) = \sum_{k=1}^K u_k \otimes \Big(v_k \sum_{i=1}^n b_i m_i\Big).$$

By our initial assumption, $\sum_{i=1}^{n} b_i m_i = 0$ in M. Therefore:

$$\sum_{i=1}^{n}\sum_{k=1}^{K}(b_i u_k)\otimes(v_k m_i) = 0 \quad \text{in } B\otimes_A M. \tag{2.2}$$

Step 3: Utilizing the A-flatness of M.
We now leverage a standard element-theoretic property of flat modules: because M is flat over A, any relation of the form $\sum e_l \otimes x_l = 0$ in $E \otimes_A M$ (for any A-module E) implies there exist $x'_s \in M$ and $a_{ls} \in A$ such that $x_l = \sum_s a_{ls} x'_s$ and $\sum_l e_l a_{ls} = 0$.

Applying this principle to equation (2.2) with $E = B$, where the terms are indexed by pairs (i,k), there exist elements $m'_s \in M$ (for $s = 1, \dots, S$) and scalars $a_{iks} \in A$ such that:

$$v_k m_i = \sum_{s=1}^{S} a_{iks} m'_s \quad \text{in } M \text{ for all } i,k, \tag{2.3}$$

$$\sum_{i=1}^{n}\sum_{k=1}^{K}(b_i u_k)a_{iks} = 0 \quad \text{in } B \text{ for all } s. \tag{2.4}$$

Step 4: Reconstructing the B-trivialization.
Consider the evaluation map $\mu_M : B\otimes_A M \to M$ given by $b\otimes m \mapsto bm$. Applying μ_M to y_i gives:

$$\mu_M(y_i) = \sum_{k=1}^{K} u_k(v_k m_i) = \Big(\sum_{k=1}^{K} u_k v_k\Big)m_i = 1\cdot m_i = m_i.$$

Substitute equation (2.3) into this expansion of m_i:

$$m_i = \sum_{k=1}^{K} u_k\Big(\sum_{s=1}^{S} a_{iks}m'_s\Big) = \sum_{s=1}^{S}\Big(\sum_{k=1}^{K} u_k a_{iks}\Big)m'_s.$$

Define $d_{is} = \sum_{k=1}^{K} u_k a_{iks} \in B$. This immediately gives our required reconstruction:

$$m_i = \sum_{s=1}^{S} d_{is} m'_s \quad \text{for all } i = 1,\dots,n.$$

Finally, we must verify that these coefficients annihilate b_i. We compute:

$$\sum_{i=1}^{n} b_i d_{is} = \sum_{i=1}^{n} b_i\Big(\sum_{k=1}^{K} u_k a_{iks}\Big) = \sum_{i=1}^{n}\sum_{k=1}^{K}(b_i u_k)a_{iks}.$$

By equation (2.4), this sum is exactly 0 in B for every s.

Thus, we have successfully trivialized the arbitrary relation $\sum_{i=1}^{n} b_i m_i = 0$ over B. By the equational criterion, M is a flat B-module.

□

Lemma 2.16 ([Sta18, Tag 092I] (1))**.** *Let $A \to B$ be a ring map such that $B\otimes_A B \to B$ is flat. If A is an absolutely flat ring, then so is B.*

Proof. First proof (the formalized version):

It follows from Lemma 2.15 immediately.

Second proof by element chasing:

Let $A = B \otimes_k B$. The multiplication map $\mu : A \to B$ is a surjective ring homomorphism. Because B is assumed to be a flat A-module, μ is a *flat epimorphism*. A fundamental property of flat epimorphisms is that their kernel, $J = \ker(\mu)$, is a *pure ideal*. The equational criterion for flatness dictates that for any element $z \in J$, there exists a "localizing" element $t \in A$ such that $\mu(t) = 1$ and $tz = 0$.

Let $\mathfrak{m}$ be the unique maximal ideal of the local ring B. We wish to show that $\mathfrak{m} = 0$. Suppose for the sake of contradiction that there exists a non-zero element $x \in \mathfrak{m}$.

Consider the element $z = x \otimes 1 - 1 \otimes x \in A$. Clearly, $\mu(z) = x - x = 0$, so $z \in J$. By the pureness of J, there exists $t \in B \otimes_k B$ such that $\mu(t) = 1$ and

$$t(x \otimes 1 - 1 \otimes x) = 0 \quad \text{in } B \otimes_k B.$$

Let $K = B/\mathfrak{m}$ be the residue field of B, and let $\pi : B \to K$ be the natural quotient map. We base-change our equation along the homomorphism $1 \otimes \pi : B \otimes_k B \to B \otimes_k K$. Because $x \in \mathfrak{m}$, we have $\pi(x) = 0$, which means $(1 \otimes \pi)(1 \otimes x) = 1 \otimes 0 = 0$.

Let $\bar{t} = (1 \otimes \pi)(t) \in B \otimes_k K$. Applying $1 \otimes \pi$ to our annihilator equation isolates the $x \otimes 1$ term:

$$\bar{t} \cdot (x \otimes 1) = 0 \quad \text{in } B \otimes_k K.$$

Since K is a field extension of k, it is a free k-module. Because tensor products preserve freeness, $N = B \otimes_k K$ is a free left B-module. The element $x \otimes 1$ represents scalar multiplication by x on N. Thus, the relation $x\bar{t} = 0$ implies that x annihilates the *content ideal* $c(\bar{t})$ of $\bar{t}$ in B (the ideal generated by the coordinates of $\bar{t}$ with respect to a basis), meaning $x \cdot c(\bar{t}) = 0$.

Now, consider the evaluation map $\mu_K : B \otimes_k K \to K$ given by $b \otimes v \mapsto \pi(b)v$. This evaluates $\bar{t}$ at the central point of the local ring:

$$\mu_K(\bar{t}) = \mu_K((1 \otimes \pi)(t)) = \pi(\mu(t)) = \pi(1) = 1.$$

Since $\mu_K(\bar{t}) = 1 \neq 0$, the coordinates of $\bar{t}$ cannot all map to zero under π. Thus, they cannot all lie in $\mathfrak{m}$. This implies the content ideal $c(\bar{t})$ is not contained in $\mathfrak{m}$.

Because B is a local ring, the only ideal not contained in the unique maximal ideal is the unit ideal itself. Therefore, $c(\bar{t}) = B$.

Substituting this into our annihilator relation yields $x \cdot B = 0$, which forces $x = 0$. This directly contradicts our initial assumption that $x \neq 0$. We conclude that $\mathfrak{m} = 0$, meaning the local ring B has only the trivial maximal ideal. Therefore, B is a field.

□

Lemma 2.17 ([Sta18, Tag 0905])**.** *Let X be a spectral space in which every quasi-compact open subset is closed. Then X is profinite.*

Proof. It suffices to show that X is Hausdorff and totally disconnected. Let $x \neq y$ be in X. Since X is quasi-sober and pre-spectral, we may assume that there exists a quasi-compact open U with $x \in U$ and $y \notin U$. By assumption, its complement is open and X is Hausdorff. This argument also shows that x and y lie in distinct connected components, so X is totally disconnected. □

Lemma 2.18 ([Sta18, Tag 092F] (2) => (3) (4))**.** *If A is an absolutely flat ring, then A is reduced and every prime is maximal.*

Proof. It suffices to show that for every $f \in A$, there exists an idempotent element $e \in A$ such that $(f) = (e)$. Indeed, if $f^n = 0$ for some n, then $e = e^n \in (f^n) = 0$. Hence $f = 0$. For the second claim observe that every quasi-compact open $D(f) = D(e)$ is closed. Hence the claim follows from 2.17 and the fact that every quasi-compact open of $\mathrm{Spec}(A)$ is a finite union of basic opens.

Now let $f \in A$ be any element. Since (f) is pure by assumption it is nilpotent by `Ideal.isIdempotentElem_of_pur` and every idempotent finitely generated ideal is generated by an idempotent element by `Ideal.isIdempotentElem_iff` ☐

Lemma 2.19. *If A is reduced and every prime is maximal, then every local ring of A is a field.*

Proof. Localizations of reduced rings are reduced and reduced, local rings of Krull dimension ≤ 0 are fields. ☐

2.2 Weak dimension

Currently, mathlib does not have Tor so we give a non-standard, but equivalent definition which suffices for our purposes.

Definition 2.20. Let A be a ring. We say it is of *weak dimension* ≤ 1 if every finitely generated ideal of A is flat over A.

Lemma 2.21 ([Sta18, Tag 092S], (2) => (4))**.** *Let A be a ring of weak dimension ≤ 1 and M a flat A-module. Then any submodule of M is flat.*

Proof. It suffices to show that for any finitely generated ideal I, the linear map $I \otimes_R N \to N$ is injective. For this, consider the following commutative diagram:

$$\begin{tikzcd} I \otimes_R N \arrow[r] \arrow[d] & N \arrow[d] \\ I \otimes_R M \arrow[r] & M \end{tikzcd}.$$

The left vertical map is injective, because I is flat by assumption on R. The bottom horizontal map is injective, because M is flat. Hence $I \otimes_R N \to N \to M$ is injective, from which the claim follows. ☐

Corollary 2.22 ([Sta18, Tag 092S], (1) => (2))**.** *If A is of weak-dimension ≤ 1, then every ideal $I \subseteq A$ is flat.*

Proof. Immediate consequence of Lemma 2.21. ☐

Lemma 2.23 ([Sta18, Tag 092E])**.** *Let A be of weak dimension ≤ 1 and B a weakly étale A-algebra. Then B is of weak dimension ≤ 1.*

Proof. Let M be a flat B-module and $N \subseteq M$ a submodule. By Lemma 2.15 it suffices to show that N is A-flat. Since B is A-flat, M is also A-flat and hence its submodule N is A-flat by Lemma 2.21. ☐

Lemma 2.24. *Let $(A_i)_{i \in I}$ be a family of rings and for every i an A_i-module M_i. Let N be a $\prod_{i \in I} A_i$-submodule of $\prod_{i \in I} M_i$. Then $N \subseteq \prod_{i \in I} N_i$. If N is finitely generated, equality holds.*

Proof. TBA. ☐

Lemma 2.25 ([Sta18, Tag 092T])**.** *Let $(A_i)_{i\in I}$ be a family of valuation rings. Then $\prod_{i\in I} A_i$ is of weak dimension ≤ 1.*

Proof. Let $I \subseteq \prod_{i\in I} A_i$ be a finitely generated ideal. Let $I_i \subseteq A_i$ be the image of I in A_i. By Lemma 2.24, I is the product of the I_i. Since each A_i is a valuation ring, there exists $f_i \in A_i$ such that $I_i = (f_i)$. Hence $I = (f)$ where $f = (f_i)$. Let $e \in A$ be the idempotent element defined by

$$e_i = \begin{cases} 0 & f_i = 0 \\ 1 & f_i \neq 0 \end{cases}$$

and let g be the element defined by

$$g_i = \begin{cases} 1 & f_i = 0 \\ f_i & f_i \neq 0 \end{cases}.$$

Since every A_i is a domain, g is a non-zerodivisor and $f = ge$. In particular, the A-linear map $(e) \to (ge) = (f)$ is an isomorphism: It is surjective by construction and injective because g is a nonzerodivisor. Since e is idempotent, we have $A = (e) \oplus (1-e)$ as A-modules, so (e) is projective. □

Lemma 2.26 ([Sta18, Tag 092V])**.** *Let A be of weak dimension ≤ 1. Let B be a flat A-algebra such that $A \to B$ is injective and an epimorphism of rings. Then A is integrally closed in B.*

Proof. Let $x \in B$ be integral over A and $A' = A[x]$. The map $A \to A'$ is injective and finite, so to show $A = A'$ it suffices to check that $A \to A'$ is an epimorphism. By Lemma 2.21, A' is flat over A. Hence the composition

$$A' \otimes_A A' \to B \otimes_A B \to B$$

is injective. □

Lemma 2.27 ([Sta18, Tag 092U])**.** *Let A be a domain and L an algebraic extension of* $\mathrm{Frac}(A)$*. If A is integrally closed in L, then there exists a cartesian diagram of rings*

$$\begin{tikzcd} A \arrow[r] \arrow[d] & L \arrow[d] \\ S \arrow[r] & T \end{tikzcd}$$

with S of weak dimension ≤ 1 and $S \to T$ a flat, injective epimorphism of rings.

Proof. TBA. □

Lemma 2.28 ([Sta18, Tag 092W])**.** *Let A be a domain and B a weakly étale A-algebra. Let L be an algebraic extension of* $\mathrm{Frac}(A)$ *and assume that A is integrally closed in L. Then B is integrally closed in $B \otimes_A L$.*

Proof. □

2.3 Fields

Lemma 2.29 ([Sta18, Tag 092P])**.** *Let L/K be an extension of fields. If $L \otimes_K L \to L$ is flat, then L is an algebraic separable extension of K.*

Proof. TBA. $\square$

Theorem 2.30 ([Sta18, Tag 092Q], (2) => (3) and last part)**.** *Let K be a field. Suppose that $K \to B$ is weakly étale, then*

1. *every finitely generated K-subalgebra of B is étale over K,*
2. *B is a filtered colimit of étale K-algebras, i.e. B is ind-étale.*

Proof. It suffices to show the first statement. A field is absolutely flat ring by Lemma 2.11, hence B is a absolutely flat ring by Lemma 2.16. And B is reduced and every local ring is a field, by Lemma 2.18 and Lemma 2.19.

Let $q \subset B$ be a prime. The ring map $B \to B_q$ is weakly étale by Lemma 2.10, hence B_q is weakly étale over K (Lemma 2.12). Thus B_q is a separable algebraic extension of K by Lemma 2.29.

Let $K \subset A \subset B$ be a finitely generated K-sub algebra. We will show that A is étale over K which will finish the proof of the lemma.

TBA.

Then every minimal prime $p \subset A$ is the image of a prime q of B, see Algebra, Lemma 00FK. Thus (p) as a subfield of Bq= (q) is separable algebraic over K. Hence every generic point of Spec(A) is closed (Algebra, Lemma 00GA). Thus dim(A)=0. Then A is the product of its local rings, e.g., by Algebra, Proposition 00KJ. Moreover, since A is reduced, all local rings are equal to their residue fields which are finite separable over K. This means that A is étale over K by Algebra, Lemma 00U3 and finishes the proof. $\square$

2.4 Local rings

Lemma 2.31. *Let A be a ring, let $f : R \to R'$, $g : S \to S'$ be two ring maps of A-algebras. Then the kernel of $R \otimes_A S \to R' \otimes_A S'$ is generated by $\ker f \otimes_A S$ and $R \otimes_A \ker g$.*

Proof. Omitted. $\square$

Lemma 2.32. *Let k be a field, A be a local k-algebra of dimension zero, i.e. there is a unique prime ideal in A. Suppose that the residue field of A is identified with k. Then $A \otimes_k A$ is a local k algebra of dimension zero.*

Proof. The unique prime ideal of A is the nilradical N of A. By Lemma 2.31, the kernel of $A \otimes_k A \to k = A/N$ is generated by $N \otimes_k A$ and $A \otimes_k N$, which are all consisiting of nilpotent elements. Thus the nilpotent ideal of $A \otimes_k A$ is also the maximal ideal. This finishes the proof. $\square$

Lemma 2.33. *Let $A \to B$ be a ring map. For all $\mathfrak{p} \subset A$, there is a unique prime $\mathfrak{q} \subset B$ lying over $\mathfrak{p}$ and $\kappa(\mathfrak{p}) = \kappa(\mathfrak{q})$. Then $B \otimes_A B \to B$ is bijective on spectra.*

Proof. It suffices to show that, for every prime p of A, there is exactly one prime ideal q' of $B \otimes_A B$ lying over p. To detect this, we may assume A is a field by replacing A with $\kappa(\mathfrak{p})$, B with $\kappa(\mathfrak{p}) \otimes_A B$ which is a local ring. Now the goal follows from Lemma 2.32. $\square$

Lemma 2.34. *Let $A \to B$ be a ring map that is bijective on spectra, as well as surjective and flat, then it is an isomorphism.*

Proof. Recall that a *pure* ideal I in A is an ideal such that A/I is flat. (Use `Ideal.Pure`.) In our case, let I be the kernel of the map $A \to B$. Then I is pure and has empty vanishing locus. But we know that pure ideals are determined by their vanishing locus (`Ideal.zeroLocus_inj_of_pure`) and zero ideal is also a pure ideal having the same vanishing locus with I, thus $I = 0$. □

Lemma 2.35 ([Sta18, Tag 04VN])**.** *Let $A \to B$ be a ring map. The following are equivalent:*

1. *$A \to B$ is an epimorphism;*
2. *the two ring maps $B \to B \otimes_A B$ are equal;*
3. *either of the ring maps $B \to B \otimes_A B$ is an isomorphism, and*
4. *the ring map $B \otimes_A B \to B$ is an isomorphism.*

Proof. Omitted. Use `Algebra.IsEpi`. □

Lemma 2.36 ([Sta18, Tag 04VU])**.** *Let $A \to B$ be a faithfully flat ring epimorphism. Then $A \to B$ is an isomorphism.*

Proof. By **??**, the map $B \to B \otimes_A B$, which is a base change of $A \to B$, is isomorphism. So is $A \to B$ by faithfully flatness. □

Lemma 2.37. *Let $A \to B$ be a weakly étale local homomorphism of local rings. For all $\mathfrak{p} \subset A$, there is a unique prime $\mathfrak{q} \subset B$ lying over $\mathfrak{p}$ and $\kappa(\mathfrak{p}) = \kappa(\mathfrak{q})$. Then $A \to B$ is an isomorphism.*

Proof. To see this, suppose that we are under the above hypothesis. This implies that $\mu : B \otimes_A B \to B$ is bijective on spectra by Lemma 2.33, as well as surjective and flat (immediately by Definition 2.8). Hence it is an isomorphism by Lemma 2.34. Together with the fact that $A \to B$ is a faithfully (by surjectivity on spectrum) flat, we can apply Lemma 2.36 to conclude $A = B$. □

Lemma 2.38. *Let I be a directed set. Let T_i, $i \in I$ be a series of totally disconnected topological spaces. Then*

$$T = \varprojlim_{i \in I} T_i$$

is also totally disconnected.

Proof. Suppose x, y are two distinct points that lives in the same connected component. Then the projection of x, y will be always fall in the same connected components, thus equal. This is a contradiction. □

Lemma 2.39. *Let B be a ind-étale algebra over some field K. If there are two different prime ideals q_1 and q_2 in B, then B has a nontrivial idempotent element e (i.i. $e^2 = e$).*

Proof. Writing B as a filtered colimit $\operatorname{colim} B_i$ of étale algebra over K. Each B_i is a product of separable field extensions of K. In particular, the spectrum of B_i's are totally disconnected. By Lemma 2.38, the spectrum of B is again totally disconnected. Thus different prime ideals q_1 and q_2 livs in different connected components. This giving the desire nontrivial idempotent element. □

Lemma 2.40. *Let B be a ind-étale algebra over some field K. If there exists a prime ideal q of B, such that the residue field of q is not K, then $\kappa(q) \otimes_A B$ has a nontrivial idempotent element.*

Proof. Writing B as a filtered colimit $\operatorname{colim} B_i$ of étale algebra over K. (Each B_i is a product of separable field extensions of K.) Let q_i be the inverse image of q in B_i, $\kappa(q_i)$ be the residue field of q_i, then K_i is a separable field extensions of K. Now $\kappa(q) = \operatorname{colim} \kappa(q_i)$ is again a separable extension of K. Then $\kappa(q) \otimes_A B = \kappa(q) \otimes_K \kappa(q)$ splits. The conclusion follows. □

Lemma 2.41. *Let R be a Henselian local ring with separably closed residue field and S a weakly étale local R-algebra with $R \to S$ a local homomorphism. Let $\mathfrak{p}$ be a prime ideal of R and L an algebraic field extension of $\kappa(\mathfrak{p})$. Then $L \otimes_R S$ has no non-trivial idempotents.*

Proof. Let R' be the integral closure of $R/\mathfrak{p}$ in L. Since $R \to R/\mathfrak{p} \to R'$ is integral, R' is local by Lemma 2.4. Moreover, the residue field $\kappa(R')$ is an algebraic extension of $\kappa(R)$ by Lemma 2.6. Since $\kappa(R)$ is separably closed, the extension is purely inseparable. Hence $R' \otimes_R S$ is local by Lemma 2.7. By Lemma 2.28 and since R' is integrally closed in L, the tensor product $R' \otimes_R S$ is integrally closed in $L \otimes_R S$. Since $R' \otimes_R S$ is local and any idempotent of $L \otimes_R S$ is integral over $R' \otimes_R S$, the latter can not have any non-trivial idemponent elements. □

Theorem 2.42 ([Sta18, Tag 097Z] (Olivier).)**.** *Let $A \to B$ be a local homomorphism of local rings. If A is henselian, the residue field of A is separably closed, and $A \to B$ is weakly étale, then $A = B$.*

Proof. It suffices to show that for all $\mathfrak{p} \subset A$ there is a unique prime $\mathfrak{q} \subset B$ lying over $\mathfrak{p}$ and $\kappa(\mathfrak{p}) = \kappa(\mathfrak{q})$ by Lemma 2.37

Note that the fibre ring $\kappa(\mathfrak{p}) \otimes_A B$ is weakly etale over $\kappa(\mathfrak{p})$ by Lemma 2.14, thus it is a colimit of étale extensions of $\kappa(\mathfrak{p})$ by Theorem 2.30. If the conclusion does not hold, at least one of the following cases is true:

1. there exists more than one prime $\mathfrak{q}$_1, $\mathfrak{q}$_2 lying over $\mathfrak{p}$;
2. if $\kappa(\mathfrak{p}) \neq \kappa(\mathfrak{q})$ for some $\mathfrak{q}$.

In the first case, by Lemma 2.39; in the second case, by Lemma 2.40, we can always find some L then $L \otimes_A B$ has a nontrivial idempotent for some (separable) algebraic field extension $L/\kappa(\mathfrak{p})$.

Now the statement follows from Lemma 2.41. □

Theorem 2.43. *Let $A \to B$ be a local homomorphism of local rings. If A is strictly henselian, and $A \to B$ is weakly étale, then $A = B$.*

Proof. This is simply Theorem 2.42 plus Lemma 1.49. □

2.5 Bijective on stalks and ind-Zariski

Lemma 2.44 ([Sta18, Tag 097F])**.** *Let $A \to B$ be a w-local ring map of w-local-rings that identifies local rings. Then $A \to B$ is ind-Zariski.*

Proof. TBA. □

Proposition 2.45 ([Sta18, Tag 097G])**.** *Let $A \to B$ be a ring map that is bijective on stalks. Then there exists a faithfully flat, ind-Zariski map $B \to B'$ such that $A \to B'$ is ind-Zariski.*

Proof. By Lemma 1.32, the induced map $A_w \to B_w$ is again bijective on stalks and hence ind-Zariski by Lemma 2.44. Since $B \to B_w$ is ind-Zariski and faithfully flat and the composition $A \to B \to B_w$ is ind-Zariski, the claim follows. □

2.6 Weakly étale and ind-étale

Theorem 2.46. *Let $R \to S$ be weakly étale. Then there exists a faithfully flat ind-étale morphism $S \to T$ such that the composition $R \to T$ is ind-étale.*

Proof. TBA. $\square$

Chapter 3

Replete categories

Let $\mathcal{C}$ be a category with limits. By $\mathrm{Ab}(\mathcal{C})$ we denote its abelian group objects.

Definition 3.1 ([BS15, Definition 3.1.1])**.** We say $\mathcal{C}$ is *replete* if epimorphisms are stable under sequential limits, i.e., if $(F_n)_{n\in\mathbb{N}}$ is an inverse system in $\mathcal{C}$ such that $F_{n+1} \to F_n$ is an epimorphism for all $n \in \mathbb{N}$, then $\lim F_n \to F_n$ is an epimorphism for all $n \in \mathbb{N}$.

The following examples are all not needed later, but are useful to gain some intuition for 3.1. The motivating example is:

Example 3.2 ([BS15, Example 3.1.4])**.** The category of sets is replete.

The category of étale sheaves is rarely replete as the following example shows.

Example 3.3 ([BS15, Example 3.1.5])**.** Let k be a field and $\bar{k}$ a separable closure of k. Then $\mathrm{Shv}(\mathrm{Spec}(k)_{\text{ét}})$ is replete if and only if $\bar{k}$ is a finite extension of k.

Example 3.4 ([BS15, Example 3.1.7])**.** The category of fpqc sheaves, i.e., sheaves on the big fpqc site, is replete.

Lemma 3.5 ([BS15, Lemma 3.1.8])**.** *Let $\mathcal{C}$ be replete and let $(F_n)_{n\in\mathbb{N}} \to (G_n)_{n\in\mathbb{N}}$ be a morphism of inverse systems in $\mathcal{C}$. If $F_n \to G_n$ and $F_{n+1} \to F_n \times_{G_n} G_{n+1}$ are epimorphisms for each $n \in \mathbb{N}$, then $\lim F_n \to \lim G_n$ is an epimorphism.*

Proposition 3.6 ([BS15, Proposition 3.1.9])**.** *If $\mathcal{C}$ is replete, countable products are exact.*

Proposition 3.7 ([BS15, Proposition 3.1.10])**.** *Let $\mathcal{C}$ be replete. If $(F_n)_{n\in\mathbb{N}}$ is an inverse system in $\mathrm{Ab}(\mathcal{C})$ with epimorphic transition maps, then $\lim F_n \cong \mathrm{R}\lim F_n$.*

3.1 Coherent objects

Definition 3.8 ([nLa25])**.** An object X of $\mathcal{C}$ is *compact* if the top element of the poset of subobjects of X is a *compact element*.

Definition 3.9 ([nLa25])**.** An object X of $\mathcal{C}$ is *stable* if for all compact Y and morphisms $Y \to X$, the fiber product $Y \times_X Y$ is compact.

Definition 3.10 ([nLa25])**.** An object X of $\mathcal{C}$ is *coherent* if it is compact and stable.

3.2 Weakly contractible categories

From now on, we assume that pullbacks in $\mathcal{C}$ preserve epimorphisms. For example this holds in abelian categories and (pre)topoi.

Definition 3.11 ([BS15, Definition 3.2.1])**.** An object F of $\mathcal{C}$ is called *weakly contractible* if every epimorphism $G \to F$ has a section. We say $\mathcal{C}$ has *enough weakly contractible objects*, if every object admits an epimorphism from a weakly contractible one.

Remark 3.12. Under our assumptions on $\mathcal{C}$, an object F is weakly contractible if and only if it is projective.

Definition 3.13 ([BS15, Definition 3.2.1])**.** We say $\mathcal{C}$ is *locally weakly contractible*, if each X in $\mathcal{C}$ admits an epimorphism from the coproduct of Y_i, where Y_i is coherent and weakly contractible.

Remark 3.14. If $\mathcal{C}$ is locally weakly contractible, it has enough contractible objects. Indeed, the coproduct of projective objects is projective.

Example 3.15. The category of sets is locally weakly contractible.

Lemma 3.16. *Let $\mathcal{C}$ have enough weakly contractible objects and $F \to G$ be a morphism. Then $F \to G$ is an epimorphism if and only if for every weakly contractible Y in $\mathcal{C}$, the induced map $F(Y) = \mathrm{Hom}(Y, F) \to \mathrm{Hom}(Y, G) = G(Y)$ is surjective.*

Proof. Choose an epimorphism $e\colon P \to G$ where P is weakly contractible and lift that to $P \to F$. $\square$

Proposition 3.17. *If $\mathcal{C}$ has enough weakly contractible objects, it is replete.*

Proof. This follows from 3.16, the fact that $\mathrm{Hom}(Y, -)$ commutes with limits and the corresponding statement in the category of sets. $\square$

Chapter 4

Pro-categories

Let $\mathcal{C}$ be a category. The category $\mathrm{Pro}(\mathcal{C})$ is the full subcategory of $\mathrm{Fun}(\mathcal{C}, \mathrm{Set})^{\mathrm{op}}$ on the functors that are cofiltered limits of representable functors. This category satisfies the following universal property:

Proposition 4.1 (Universal property of $\mathrm{Pro}(\mathcal{C})$)**.** *Let $\mathcal{D}$ be a category with cofiltered limits. The restriction functor*

$$\mathrm{Fun}(\mathrm{Pro}(\mathcal{C}), \mathcal{D}) \to \mathrm{Fun}(\mathcal{C}, \mathcal{D}),$$

admits a left-adjoint that induces an equivalence on the full subcategory of $\mathrm{Fun}(\mathrm{Pro}(\mathcal{C}), \mathcal{D})$ *on functors preserving cofiltered limits.*

Proof. The left-adjoint is the left Kan extension along the inclusion $\mathcal{C} \to \mathrm{Pro}(\mathcal{C})$. ☐

Let now $\mathcal{K}$ be a small category. The canonical functor $\mathcal{C} \to \mathrm{Pro}(\mathcal{C})$ induces a functor $\mathrm{Fun}(\mathcal{K}, \mathcal{C}) \to \mathrm{Fun}(\mathcal{K}, \mathrm{Pro}(\mathcal{C}))$. Since the right hand side admits small cofiltered limits, the universal property of $\mathrm{Pro}(\mathcal{C})$ induces a canonical functor

$$\Phi_{\mathcal{K}} \colon \mathrm{Pro}(\mathrm{Fun}(\mathcal{K}, \mathcal{C})) \longrightarrow \mathrm{Fun}(\mathcal{K}, \mathrm{Pro}(\mathcal{C})).$$

Proposition 4.2. *Suppose $\mathcal{K}$ is a finite category and every object of $\mathcal{K}$ admits no non-trivial automorphisms. Then the natural functor Φ_K induces an equivalence of categories.*

Proof. See [KS06, Theorem 6.4.3]. ☐

Example 4.3. Note that Proposition 4.2 in particular implies that if there is a finite family of morphisms $f_i \colon X_i \to X$ in $\mathrm{Pro}(C)$, there exists a diagram $\mathcal{J}$ such that $X = \lim_{j \in \mathcal{J}} Y_j$ and $X_i = \lim_{j \in \mathcal{J}} Y_{ij}$ and $f_i = \lim_{j \in \mathcal{J}} f_{ij}$ for morphisms $f_{ij} \colon Y_{ij} \to Y_j$.

Proposition 4.4. *Let $\mathcal{A}$ be a category with cofiltered limits and suppose there exists a fully faithful functor $i \colon \mathcal{C} \to \mathcal{A}$ such that $i(X)$ is cocompact for all $X \in \mathcal{C}$. Then*

$$\lim_{\mathcal{A}} \circ \mathrm{Pro}(i) \colon \mathrm{Pro}(\mathcal{C}) \to \mathcal{A}$$

is fully faithful.

Proof. See nlab. ☐

Corollary 4.5. *Let P be a property of objects on a category $\mathcal{A}$ that implies finitely presented. Denote by $\mathcal{A}_P$ (resp. $\mathcal{A}_{\mathrm{Pro}(P)}$) the full subcategory defined by P (resp. $\mathrm{Pro}(P)$). Suppose $\mathcal{A}$ has cofiltered limits, then $\lim_{\mathcal{A}}$ induces an equivalence*

$$\mathrm{Pro}(\mathcal{C}_P) \cong \mathcal{C}_{\mathrm{Pro}(P)}.$$

Proof. Follows from 4.4, because by definition $\mathcal{C}_{\mathrm{Pro}(P)}$ is the essential image of $\lim_{\mathcal{A}}(\mathrm{Pro}(\mathcal{C}_P))$. □

Note that 4.5 does not apply to $\mathcal{A} = \mathrm{Scheme}/X$ and $P =$ étale, because Scheme/X does not have all cofiltered limits. It does apply in the case of $\mathcal{A} = \mathrm{AffScheme}/X$ though or in the case $\mathrm{CRing}^{\mathrm{op}}/R$.

Recall the following definition:

Definition 4.6. A category $\mathcal{C}$ is *precoherent* if given a finite effective epimorphism family $f\colon : X_i \to X$ and a morphism $p\colon Y \to X$, there exists a finite effective epimorphism family $g\colon Y_j \to Y$ that factors through f.

Lemma 4.7. *A morphism $f\colon X \to Y$ in $S_{\text{ét}}$ is an effective epimorphism if and only if it is surjective.*

Proof. One direction follows from the fact that the fpqc topology is subcanonical. The converse holds, because étale morphisms are open: If $f\colon X \to Y$ is an étale morphism we may glue two copies of Y along the inclusion of the image of f. If f is not surjective, the two inclusions of Y into the amalgamated sum are different, but by construction they agree on the image of f. Hence f is not an epimorphism. □

Lemma 4.8. *The category of schemes is finitary extensive.*

Proof. Since the Zariski topology is subcanonical, the category of schemes embeds into the big Zariski topos, which is finitary extensive. Since the Yoneda embedding preserves finite coproducts, the claim follows. □

Lemma 4.9. *The category $X_{\text{ét}}$ is precoherent.*

Proof. The category of schemes is finitary extensive, so the same holds for $X_{\text{ét}}$. By 4.7 the effective epimorphisms in $X_{\text{ét}}$ are the surjective morphisms, hence being an effective epimorphism is stable under base change. Thus $X_{\text{ét}}$ is preregular and therefore precoherent. □

Proposition 4.10. *If $\mathcal{C}$ is precoherent, also $\mathrm{Pro}(\mathcal{C})$ is precoherent.*

4.1 Coherent topology

Any precoherent category can be equipped with the coherent coverage, where covering families are given by finite effective epimorphic families. The coherent topology is the topology generated by the coherent coverage.

Remark 4.11. One would hope that the topology on $X_{\text{ét}}$ given by jointly surjective families is the precoherent topology. Since effective epimorphisms are simply surjective maps in $X_{\text{ét}}$, this would mean that every cover in $X_{\text{ét}}$ can be refined by a finite cover. This is not true though, because $X_{\text{ét}}$ also contains non-quasi-compact schemes.

The situation is slightly better if we instead consider the affine étale site $X^{\mathrm{aff}}_{\text{ét}}$ of X, the category of affine schemes étale over X. Unfortunately, it is not clear to the authors if effective epimorphisms in $X^{\mathrm{aff}}_{\text{ét}}$ are still surjective.

Let from now on $\mathcal{C}$ be precoherent equipped with the coherent topology.

Proposition 4.12. *Let $F\colon \mathrm{Pro}(\mathcal{C})^{\mathrm{op}} \to \mathrm{Set}$ be a sheaf. Then $F|_{\mathcal{C}^{\mathrm{op}}}$ is a sheaf.*

By 4.12, the inclusion functor $\mathcal{C} \to \mathrm{Pro}(\mathcal{C})$ induces a morphism on sites $\nu\colon \mathrm{Pro}(\mathcal{C}) \to \mathcal{C}$. The direct image functor $\nu_*\colon \mathrm{Shv}(\mathrm{Pro}(\mathcal{C})) \to \mathrm{Shv}(\mathcal{C})$ agrees with the restriction $F \mapsto F|_{\mathcal{C}^{\mathrm{op}}}$.

Proposition 4.13. *Let $F\colon \mathrm{Pro}(\mathcal{C})^{\mathrm{op}} \to \mathrm{Set}$ be a presheaf. Suppose $F|_{\mathcal{C}^{\mathrm{op}}}$ is a sheaf and F preserves filtered colimits. Then F is a sheaf.*

Combining 4.1 and 4.13 yields:

Corollary 4.14. *Let F be a sheaf on $\mathcal{C}$. Then the unit $F \to \nu_*\nu^{-1}F$ is an isomorphism.*

Proof. Denote by res the restriction functor $\mathrm{PShv}(\mathrm{Pro}(\mathcal{C})) \to \mathrm{PShv}(\mathcal{C})$ and by ext the left-adjoint of 4.1. By 4.13, ext restricts to a functor $\mathrm{Shv}(\mathrm{Pro}(\mathcal{C})) \to \mathrm{Shv}(\mathcal{C})$ that is the left-adjoint of ν_*. In particular it agrees with ν^{-1} and the result follows. □

4.2 Generated topologies

The previous section established a relationship between the coherent topologies on $\mathcal{C}$ and $\mathrm{Pro}(\mathcal{C})$. In the situations we are interested in, the topologies don't necessarily agree with the coherent topologies though (see Remark 4.11).

To remedy the situation, we make the following definitions:

Definition 4.15. Let $\mathcal{U}$ be a one-hypercover in a site $\mathcal{C}$. We say $\mathcal{U}$ is *exact* if taking filtered colimits preserves the multiequalizer diagram of $\mathcal{U}$.

Example 4.16. Any finite one-hypercover is exact, because filtered colimits commute with finite limits.

Definition 4.17. Let $F\colon \mathcal{C} \to \mathcal{D}$ be a functor between sites. We say a one-hypercover $\mathcal{U}$ of an object X in $\mathcal{D}$ is *relatively pro-representable* if it is exact and $\mathcal{U}$ can be written as the componentwise cofiltered limit of one-hypercovers in $\mathcal{C}$.

Concretely, a one-hypercover $\mathcal{U} = (U_\alpha, U_{\alpha\beta\gamma})$ is relatively pro-representable if there exists

1. a cofiltered diagram $\mathcal{I}$,
2. functors $V_\alpha\colon \mathcal{I} \to \mathcal{C}$ and $V_{\alpha\beta\gamma}\colon \mathcal{I} \to \mathcal{C}$,
3. natural transformations $p_1\colon V_{\alpha\beta\gamma} \to V_\alpha$ and $p_2\colon V_{\alpha\beta\gamma} \to V_\beta$,

such that

1. $U_\alpha = \lim_i V_\alpha(i)$ and $U_{\alpha\beta\gamma} = \lim_i V_{\alpha\beta\gamma}(i)$, and
2. for every $i \in \mathcal{I}$, the induced pre-one-hypercover $(V_\alpha(i), V_{\alpha\beta\gamma}(i))$ is a one-hypercover.

Example 4.18. Suppose $\mathcal{C}$ is coherent, $\mathcal{D} = \mathrm{Pro}(\mathcal{C})$ and $F\colon \mathcal{C} \to \mathrm{Pro}(\mathcal{C})$ the inclusion. If both $\mathcal{C}$ and $\mathrm{Pro}(\mathcal{C})$ are equipped with the coherent topologies, the topology on $\mathrm{Pro}(\mathcal{C})$ is generated by relatively pro-representable one-hypercovers.

The analogue of 4.13 is:

Proposition 4.19. *Suppose the topology on $\mathcal{D}$ is generated by relatively pro-representable one-hypercovers and let P be a presheaf on $\mathcal{D}$. If $P \circ F^{\mathrm{op}}$ is a sheaf on $\mathcal{C}$ and P preserves filtered colimits, then P is a sheaf on $\mathcal{D}$.*

Proof. By the assumption, it suffices to check the sheaf condition for every relatively pro-representable one-hypercover $\mathcal{U} = \lim F(\mathcal{V}_i)$. Because P preserves filtered colimits, one computes that the sheaf condition for $\mathcal{U}$ is the filtered colimit of the sheaf conditions for $\mathcal{V}_i$. But $P \circ F^{\mathrm{op}}$ is a sheaf and filtered colimits preserve the equalizers because $\mathcal{U}$ is exact. $\square$

Chapter 5

The pro-étale topology

In this chapter we introduce the pro-étale site of a scheme and study its basic properties.

5.1 Preliminaries about the étale topology

Before going to the pro-étale case, we recall a few preliminary results about the étale site that we will need in the sequel. Let X be a scheme. By $X_{\text{ét}}$ we denote the étale site of X and by $X_{\text{ét}}^{\text{aff}}$ the affine étale site. $X_{\text{ét}}^{\text{aff}}$ consists of affine schemes étale over X.

Lemma 5.1. *The inclusion functor $X_{\text{ét}}^{\text{aff}} \to X_{\text{ét}}$ is cover dense, i.e. for every Y in $X_{\text{ét}}$ there exists a cover $\{U_i \to Y\}$ with U_i in $X_{\text{ét}}^{\text{aff}}$.*

Proof. This is immediate, because open immersions are étale and étale is stable under composition. □

5.2 Weakly étale morphisms of schemes

Definition 5.2 (Weakly étale morphisms, [BS15, Definition 4.1.1])**.** A map $f : Y \to X$ is *weakly étale* if f is flat and $\Delta_f : Y \to Y \times_X Y$ is flat.

Lemma 5.3 (Local property of weakly étale morphisms)**.** *$f : Y \to X$ is weakly étale if and only if for every affine open U of Y and every affine open V of X with $f(U) \subset V$, the induced ring homomorphism $\Gamma(X, V) \to \Gamma(Y, U)$ is weakly étale.*

Proof. Weakly étale morphisms of schemes are local on source and on target for the Zariski topology, so the property descends to affine opens. There it matches the ring-theoretic definition via the characterisation of weakly étale algebras (Definition 2.8). □

Lemma 5.4. *Etale morphisms are weakly étale.*

Proof. Any étale morphism is flat and its diagonal is an open immersion, hence flat. □

Lemma 5.5 ([BS15, Lemma 4.1.6])**.** *Weakly étale is stable under composition and base change.*

Proof. This follows because flat is stable under composition and base change. □

5.3 The site

Before we define the pro-étale site, we recall a few standard definitions.

Definition 5.6 (Quasi-compact cover)**.** A jointly surjective family $(f_i : Y_i \to X)_{i\in I}$ of morphisms of schemes is *quasi-compact* if for every affine open U of X there exist quasi-compact opens V_i in Y_i such that $U = \bigcup_{i\in I} f_i(V_i)$.

Lemma 5.7. *Let $(f_i : Y_i \to X)_{i\in I}$ be a jointly-surjective family of morphisms of schemes. If f_i is flat and locally of finite presentation for every $i \in I$, the family is quasi-compact.*

Proof. In this case, every f_i is an open map, from which the claim easily follows. □

Definition 5.8. Let $\mathcal{P}$ be a morphism property of schemes and X be a scheme. The *small $\mathcal{P}$-site of X* is the category of X-schemes with structure morphism satisfying $\mathcal{P}$ and with covers given by quasi-compact, jointly surjective families of morphisms satisfying $\mathcal{P}$.

If we let $\mathcal{P}$ to be flat morphisms of schemes, we obtain the fpqc site:

Definition 5.9 (The fpqc site)**.** Let X be a scheme. The *fqpc site of X*, denoted by X_{fpqc}, is the $\mathcal{P}$-site of X with $\mathcal{P}$ = flat.

For us, the main relevant property of the fqpc site is that it is subcanonical:

Theorem 5.10. *The fpqc site is subcanonical, i.e., every representable presheaf is a sheaf.*

Analogous to the familiar criterion for fpqc sheaves, we have a generalization for the small $\mathcal{P}$-site of a scheme.

Proposition 5.11. *Let F be a presheaf on the small $\mathcal{P}$-site. Then F is a sheaf in the if and only if it is a sheaf in the Zariski topology and satisfies the sheaf property for $\{V \to U\}$ with V, U affine and $V \to U$ satisfying $\mathcal{P}$.*

We can now define the main object of this chapter:

Definition 5.12 (The pro-étale site, [BS15, Definition 4.1.1])**.** Let X be a scheme. The *pro-étale site of X*, denoted by $X_{\text{proét}}$, is the $\mathcal{P}$-site of X with $\mathcal{P}$ = weakly étale.

Remark 5.13. Despite the name proétale, Definition 5.12 does not mention pro-étale morphisms. The name is justified by Theorem 2.46 (see also Proposition 5.18).

Proposition 5.14 ([BS15, Lemma 4.2.4])**.** *The site $X_{\text{proét}}$ is subcanonical.*

Proof. Follows from 5.10, since every pro-étale sheaf is an fpqc-sheaf. □

5.3.1 Affine covers

Definition 5.15 ([BS15, Definition 4.2.1])**.** An object U of $X_{\text{proét}}$ is called a *pro-étale affine* if it can be written as $U = \lim U_i$ over a small cofiltered diagram $i \mapsto U_i$ of affine schemes étale over X. The subcategory of pro-étale affines in $X_{\text{proét}}$ is denoted by $X^{\text{aff}}_{\text{proét}}$.

Lemma 5.16 ([BS15, Remark 4.2.3])**.** *The category $X^{\text{aff}}_{\text{proét}}$ admits limits indexed by a connected diagram and the forgetful functor to* Sch$/X$ *preserves them. If X is affine, $X^{\text{aff}}_{\text{proét}}$ has all small limits.*

Definition 5.17. Let X be affine. The *affine pro-étale site* of X is the category $X^{\mathrm{aff}}_{\mathrm{pro\acute{e}t}}$. The covering families are the quasi-compact covers.

In other words, $X^{\mathrm{aff}}_{\mathrm{pro\acute{e}t}}$ carries the topology induced by the inclusion functor $X^{\mathrm{aff}}_{\mathrm{pro\acute{e}t}} \to X_{\mathrm{pro\acute{e}t}}$. Although the objects of $X_{\mathrm{pro\acute{e}t}}$ are weakly étale X-schemes, by Theorem 2.46 the topology of $X_{\mathrm{pro\acute{e}t}}$ is still generated by objects in $X^{\mathrm{aff}}_{\mathrm{pro\acute{e}t}}$:

Proposition 5.18 ([BS15, Lemma 4.2.4])**.** *The inclusion $X^{\mathrm{aff}}_{\mathrm{pro\acute{e}t}} \to X_{\mathrm{pro\acute{e}t}}$ is cover dense, i.e. for every object Y of $X_{\mathrm{pro\acute{e}t}}$, there exists a cover $\{U_i \to Y\}_{i\in I}$ with U_i in $X^{\mathrm{aff}}_{\mathrm{pro\acute{e}t}}$.*

Proof. Let $f\colon Y \to X$ be a weakly étale morphism and let $\{U_i \to X\}$ be an affine open cover of X. If we prove the lemma for each U_i and the weakly étale morphism $f^{-1}(U_i) \to U_i$, the lemma follows for f by taking the joint cover. Thus we may assume that $X = \mathrm{Spec}(A)$ is affine.

Now let $\{U_i \to Y\}$ be an affine open cover. Again, if we prove the lemma for each U_i and the composition $U_i \to Y \to X$, the claim follows by joining the covers. Hence we may assume that $Y = \mathrm{Spec}(B)$ is affine.

We now have $A \to B$ weakly étale, so by Theorem 2.46 there exists an ind-étale and faithfully flat $B \to B'$ such that $A \to B \to B'$ is ind-étale. The cover $\{\mathrm{Spec}(B') \to \mathrm{Spec}(B)\}$ concludes the proof. □

Corollary 5.19. *Restriction along the inclusion $X^{\mathrm{aff}}_{\mathrm{pro\acute{e}t}} \to X_{\mathrm{pro\acute{e}t}}$ induces an equivalence of categories* $\mathrm{Shv}(X_{\mathrm{pro\acute{e}t}}) \to \mathrm{Shv}(X^{\mathrm{aff}}_{\mathrm{pro\acute{e}t}})$.

Proof. The functor $X^{\mathrm{aff}}_{\mathrm{pro\acute{e}t}} \to X_{\mathrm{pro\acute{e}t}}$ is fully faithful and makes $X^{\mathrm{aff}}_{\mathrm{pro\acute{e}t}}$ a dense subsite of $X_{\mathrm{pro\acute{e}t}}$ by Proposition 5.18. Thus the claim follows from the comparison lemma for sites. □

Corollary 5.20. *Restriction along the inclusion $X^{\mathrm{aff}}_{\mathrm{pro\acute{e}t}} \to X_{\mathrm{pro\acute{e}t}}$ commutes with sheafification.*

Proof. This follows immediately from 5.19. □

We can also compare $X^{\mathrm{aff}}_{\mathrm{pro\acute{e}t}}$ to $X^{\mathrm{aff}}_{\mathrm{\acute{e}t}}$.

Proposition 5.21. *The canonical functor $X^{\mathrm{aff}}_{\mathrm{\acute{e}t}} \to X^{\mathrm{aff}}_{\mathrm{pro\acute{e}t}}$ induces an equivalence of categories* $\mathrm{Pro}(X^{\mathrm{aff}}_{\mathrm{\acute{e}t}}) \to X^{\mathrm{aff}}_{\mathrm{pro\acute{e}t}}$.

Proof. By Proposition 4.4, it suffices to show that for every U in $X^{\mathrm{aff}}_{\mathrm{\acute{e}t}}$, the image of U in $X^{\mathrm{aff}}_{\mathrm{pro\acute{e}t}}$ is co-finitely-presentable. This follows from the fact that étale morphisms are finitely presented. □

5.3.2 Repleteness

An important property of the pro-étale site is the following:

Proposition 5.22 ([BS15, Proposition 4.2.8])**.** *The category of sheaves on $X_{\mathrm{pro\acute{e}t}}$ has enough weakly contractible objects.*

Proof. TBA. □

Corollary 5.23. *The category of sheaves on $X_{\mathrm{pro\acute{e}t}}$ is replete.*

5.4 Comparison with étale cohomology

Let X be a scheme. Since an étale map is also weakly étale by 5.4 we may define:

Definition 5.24. We denote by $\nu = \nu_X$ the morphism of sites $X_{\text{proét}} \to X_{\text{ét}}$ induced by the functor from $X_{\text{ét}}$ to $X_{\text{proét}}$ given by $(U \to X) \mapsto (U \to X)$.

Lemma 5.25. *The morphism of sites ν is continuous.*

Proof. This follows from the definitions, because every étale covering is a quasi-compact covering by 5.7. □

We denote by $\nu_p \colon \mathrm{PreShv}(X_{\text{proét}}) \to \mathrm{PreShv}(X_{\text{ét}})$ and $\nu^p \colon \mathrm{PreShv}(X_{\text{ét}}) \to \mathrm{PreShv}(X_{\text{proét}})$ the pushforward and pullback functors on the level of presheafs.

Lemma 5.26. *Let F be a presheaf on $X_{\text{ét}}$. Then $(\nu^p F)|_{X^{\text{aff}}_{\text{proét}}}$ preserves filtered colimits.*

Proof. Let $U = \lim_\lambda U_\lambda$ be a cofiltered limit. By definition of $X^{\text{aff}}_{\text{proét}}$ and because pro-pro-étale is pro-étale, we may assume that the U_λ are étale over X. Then

$$(\nu^p F)|_{X^{\text{aff}}_{\text{proét}}}(\lim U_\lambda) = \operatorname{colim}_\mu F(V_\mu),$$

where the V_μ run over factorisations $U = \lim U_\lambda \to V_\mu \to X$ where $V_\mu \to X$ is étale. Since the U_λ are locally of finite presentation over X, they are finitely presented in the categorical sense. Hence, for every such factorisation there exists a λ such that $U = \lim U_\lambda \to V_\mu$ factors via U_λ. This shows that the U_λ lie cofinal in all V_μ, so we obtain

$$\operatorname{colim}_\mu F(V_\mu) = \operatorname{colim}_\lambda F(U_\lambda) = \operatorname{colim}_\lambda (\nu^p F)|_{X^{\text{aff}}_{\text{proét}}}(U_\lambda).$$

□

Lemma 5.27. *The topology on $X^{\text{aff}}_{\text{proét}}$ is generated by relatively pro-representable one-hypercovers with respect to the inclusion $X^{\text{aff}}_{\text{ét}} \to X^{\text{aff}}_{\text{proét}}$.*

Proof. The topology on $X^{\text{aff}}_{\text{proét}}$ is generated by surjections $V \to U$ in $X^{\text{aff}}_{\text{proét}}$ and finite standard Zariski coverings. Both of these can be relatively pro-represented by Proposition 4.2 (see also Example 4.3). □

Further denote by $\nu_* \colon \mathrm{Shv}(X_{\text{proét}}) \to \mathrm{Shv}(X_{\text{ét}})$ and $\nu^* \colon \mathrm{Shv}(X_{\text{ét}}) \to \mathrm{Shv}(X_{\text{proét}})$ the corresponding pushforward and pullback functors. Since ν^* is exact, we have $R\nu^* = \nu^*$ by abuse of notation and we simply write ν^* everywhere. In contrast to [BS15], we write $R\nu_*$ for the derived functor of ν_*.

Proposition 5.28. *Let F be a sheaf on $X_{\text{ét}}$. Then $(\nu^p F)|_{X^{\text{aff}}_{\text{proét}}}$ is a sheaf. In particular, $(\nu^* F)|_{X^{\text{aff}}_{\text{proét}}} \cong (\nu^p F)|_{X^{\text{aff}}_{\text{proét}}}$.*

Proof. By Lemma 5.27 and Lemma 5.26, this follows from Proposition 4.19. Hence we have the following chain of isomorphisms

$$(\nu^* F)|_{X^{\text{aff}}_{\text{proét}}} \cong (\nu^p F)^\#|_{X^{\text{aff}}_{\text{proét}}} \cong ((\nu^p F)|_{X^{\text{aff}}_{\text{proét}}})^\# \cong (\nu^p F)|_{X^{\text{aff}}_{\text{proét}}}.$$

The second isomorphism is Corollary 5.20 and the last is the first part of this proposition. □

Corollary 5.29 ([BS15, Lemma 5.1.1])**.** *Let F be a sheaf on $X_{\text{ét}}$ and $U = \lim_i U_i$ in $X^{\text{aff}}_{\text{proét}}$. Then $\nu^* F(U) = \operatorname{colim}_i F(U_i)$.*

Proof. By 5.28 we have $(\nu^* F)|_{X^{\text{aff}}_{\text{proét}}} \cong (\nu^p F)|_{X^{\text{aff}}_{\text{proét}}}$. The claim now follows from 5.26. □

Corollary 5.30. *For any $F \in \operatorname{Shv}(X_{\text{ét}})$, the unit $F \to \nu_* \nu^* F$ is an isomorphism.*

Proof. By 5.1, we may check this on any affine U étale over X. Since $U = \lim U$, the formula of 5.29 shows the desired isomorphism. □

Corollary 5.31 ([BS15, Lemma 5.1.2, fully faithful part])**.** *ν^* is fully faithful.*

Proof. Follows formally from 5.30. □

Definition 5.32. A sheaf F in $\operatorname{Shv}(X_{\text{proét}})$ is called *classical* if it lies in the essential image of ν^*.

We can recognize the classical sheafs among all pro-étale sheafs by checking if they preserve affine filtered colimits:

Corollary 5.33 ([BS15, Lemma 5.1.2, essential image part])**.** *A sheaf F in $\operatorname{Shv}(X_{\text{proét}})$ is classical if and only if for any $U \in X^{\text{aff}}_{\text{proét}}$ with presentation $U = \lim U_i$, it holds that $F(U) = \operatorname{colim} F(U_i)$.*

Proof. The only if part follows directly from Corollary 5.29. Conversely, let F be in $\operatorname{Shv}(X_{\text{proét}})$ and suppose F satisfies the formula in the statement. Then $\nu^* \nu_* F \to F$ is an isomorphism. Indeed, by Corollary 5.19, it suffices to check this after restricting to $X^{\text{aff}}_{\text{proét}}$. Let $U = \lim U_i$ be in $X^{\text{aff}}_{\text{proét}}$. Then again by Corollary 5.29

$$\nu^* \nu_* F(U) = \operatorname{colim} \nu_* F(U_i) = \operatorname{colim} F(U_i) = F(U) \qquad \square.$$

Lemma 5.34 ([BS15, Lemma 5.1.6])**.** *Let K be in $D^+(X_{\text{ét}})$ and $U = \lim_i U_i$ in $X^{\text{aff}}_{\text{proét}}$. Then $R\Gamma(U, \nu^* K) = \operatorname{colim}_i R\Gamma(U_i, K)$.*

Proof. TBA. □

Proposition 5.35 ([BS15, Corollary 5.1.6])**.** *For any K in $D^+(X_{\text{ét}})$ the unit $K \to R\nu_* \nu^* K$ is an isomorphism.*

Proof. TBA. □

Corollary 5.36. *Let K be in $D^+(X_{\text{ét}})$. Then*

$$R\Gamma(X_{\text{ét}}, K) \cong R\Gamma(X_{\text{proét}}, \nu^* K).$$

Proof. By 5.35, we have $R\Gamma(X_{\text{ét}}, K) \cong R\Gamma(X_{\text{ét}}, R\nu_* \nu^* K)$. Since ν_* maps injective sheaves to acyclic ones by general theory, we have $R\Gamma(X_{\text{ét}}, -) \circ \nu_* = R(\Gamma(X_{\text{ét}}, -) \circ \nu_*) = R\Gamma(X_{\text{proét}}, -)$. So the claim follows. □

Definition 5.37 ([Jan88, §3])**.** Denote by $\Gamma_{\text{cont}}(X, -) \colon \operatorname{Ab}(X_{\text{ét}})^{\mathbb{N}} \to \operatorname{Ab}$ the functor

$$(F_n)_{n \in \mathbb{N}} \mapsto \Gamma\left(X, \lim F_n\right).$$

For an inverse system $(F_n)_{n\in\mathbb{N}}$, we call the cohomology groups $H^i \mathrm{R}\Gamma_{\text{cont}}\left(X, -\right)\left((F_n)_{n\in\mathbb{N}}\right)$, denoted by $H^i_{\text{cont}}(X_{\text{ét}}, (F_n)_{n\in\mathbb{N}})$, the *continuous étale cohomology* of X with coefficients in $(F_n)_{n\in\mathbb{N}}$.

Theorem 5.38 ([BS15, Proposition 5.6.2]). *Let $(F_n)_{n\in\mathbb{N}}$ be an inverse system of abelian sheaves on $X_{\text{ét}}$ with epimorphic transition maps. Then there exists a canonical identification*

$$R\Gamma_{cont}(X,(F_n)_n) \cong R\Gamma(X_{\text{proét}}, \lim \nu^* F_n).$$

In particular, for every $i \geq 0$ we have

$$H^i_{\text{cont}}(X_{\text{ét}},(F_n)_n) \cong H^i(X_{\text{proét}}, \lim \nu^* F_n).$$

Proof. We repeatedly use the fact that Γ commutes with inverse limits both on $X_{\text{ét}}$ and $X_{\text{proét}}$. Then we have

$$\begin{aligned}
R\Gamma_{\text{cont}}(X,(F_n)_n) &\cong R\left(\Gamma(X_{\text{ét}},-)\circ\lim\right)((F_n)_n)\\
&\cong R\left(\lim\circ\Gamma(X_{\text{ét}},-)\right)((F_n)_n)\\
&\cong R\lim(R\Gamma(X_{\text{ét}},F_n))\\
&\cong R\lim(R\Gamma(X_{\text{proét}},\nu^*F_n))\\
&\cong R\left(\lim\circ\Gamma(X_{\text{proét}},-)\right)((\nu^*F_n)_n)\\
&\cong R\left(\Gamma(X_{\text{proét}},-)\circ\lim\right)((\nu^*F_n)_n)\\
&\cong R\Gamma(X_{\text{proét}},R\lim\nu^*F_n)\\
&\cong R\Gamma(X_{\text{proét}},\lim\nu^*F_n)
\end{aligned}$$

The fourth isomorphism is 5.36 and the last is 3.7. ∎

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